

Status Externalities in Education and Low Birth Rates in Korea[†]

By SEONGEUN KIM, MICHÈLE TERTILT, AND MINCHUL YUM*

South Koreans appear to be preoccupied with their offspring's education and also have the lowest total fertility rate in the world. We propose a novel theory with status externalities and endogenous fertility connecting these facts, motivated by novel empirical evidence on private education spillovers. Using a quantitative model calibrated to Korea, we find that fertility would be 28 percent higher without the externality. We explore the effects of government policy: a pro-natal transfer or an education tax can increase fertility and reduce education spending. An education tax of 22 percent together with moderate pro-natal transfers maximizes the current generation's welfare. (JEL D31, D62, E24, I21, I28, J13, O40)

South Korea (henceforth, Korea) has an extremely low total fertility rate. For almost two decades now, Korea's fertility rate has been among the lowest in the world (reaching a record low of 0.78 in 2022), below that of other low-fertility countries such as Germany or Italy. The Korean government has expressed concerns over low birth rates, as the latter imply rapid population aging and pose a considerable challenge to the public pension system. Starting in 2006, the government launched a billion-dollar program to reverse the decline. In 2020, US\$37 billion (2.1 percent of GDP) was spent on policies aimed at boosting fertility.¹

Another notable feature of Korean society is that children's education is very highly valued by parents. This preoccupation with education is sometimes called "education fever," echoing the title of a popular book by Michael Seth (2002). Many teenagers attend math and English classes in private education institutes called *hagwons*, often as late as midnight. Others, meanwhile, spend numerous hours each week with a private tutor. Participation rates in after-school programs are around

*Kim: Sejong University (email: seongeun.kim@sejong.ac.kr); Tertilt: University of Mannheim (email: tertilt@uni-mannheim.de); Yum: University of Southampton (email: m.yum@soton.ac.uk). Emi Nakamura was the coeditor for this article. We thank four anonymous referees and the editors for their constructive comments that have improved the paper substantially. We also thank Mark Bils, Yongsung Chang, David de la Croix, Ori Heffetz, Duk Gyoo Kim, Sang Yoon (Tim) Lee, Christoph Rothe, Carsten Trenkler, Gianluca Violante, Lingwei Wu, and participants at numerous seminars and conferences for their helpful feedback. Yiwei Yin and Mingjie Xu provided excellent research assistance. We gratefully acknowledge financial support from the German Research Foundation (through the CRC-TR-224 project A3 and the Gottfried Wilhelm Leibniz-Prize).

[†]Go to <https://doi.org/10.1257/aer.20220583> to visit the article page for additional materials and author disclosure statement(s).

¹Source: "The Fourth Basic Plan on Low Fertility and Aging Society," published by the Korean Presidential Committee on Aging Society and Population Policy in 2020 (available only in Korean at <https://www.betterfuture.go.kr>).

75 percent. These private education investments are so expensive that, on average, an individual family spends as much as 9.2 percent of their income per child on education (even though most children attend public schools).

In this paper, we propose a new mechanism that connects high education spending with low birth rates. The novel ingredient is a status externality in which parents value the education of their children *relative* to the education of other children. The concern for status seems particularly relevant in Korea and East Asia more generally.² Extremely low fertility rates coupled with high education spending is a feature of other East Asian countries as well, such as in China, Hong Kong, Singapore, and Taiwan. While our analysis relies on data from Korea, we believe that the mechanism and policy implications may apply throughout East Asia.

We use micro data from Korea to explore whether spillovers in private education spending across families indeed exist. Using regional variation in the change of late-night curfews on *hagwons* as instruments, we find that lower spending on private education among relatively rich families lowered private education expenditures of socioeconomically disadvantaged (i.e., low-income or low-education) families. Since much fewer children from these relatively poorer families attend *hagwons* late at night and thus are less likely to be directly affected by the late-night curfew changes, our evidence points to the existence of spillovers across families.

Motivated by this evidence, we incorporate status externalities in education into a quantitative overlapping generations model with endogenous fertility. The model features heterogeneous agents, where potential parents choose the number of children and how much to invest in the latter's education, incorporating a quantity-quality trade-off. The children's human capital is a function of parental spending on education and a stochastic learning ability that captures parental background as well as luck. Agents differ in their human capital—largely determined by their parents—as well as an exogenous preference shock. We model the status externality by assuming that parents derive utility from the human capital of their children relative to the human capital of other people's children. Given the status externality, the key object of the equilibrium is the distribution of human capital, which is endogenously determined as a fixed point.

We calibrate our model to Korea, using data from the Korean Labor and Income Panel Study (KLIPS). We use the calibrated model to understand fertility choices and education investments in Korea and to conduct policy experiments. In particular, we use our reduced-form spillover estimate to discipline the degree of externality in the model. The first finding from the calibrated model is that the status externality plays an important role in fertility decisions: in its absence, fertility increases by 28 percent. The reason is that the status externality drives up education investments, which makes children costly and induces parents to have fewer offspring. This channel especially affects the poor. Indeed, childlessness falls from 5 percent to less than 1 percent in the lowest-income quintile when the status externality is eliminated. Removing the externality therefore changes the

²For example, Jinkins (2016) and Podoshen, Li, and Zhang (2011) find that Chinese consumers care more about peer beliefs and conspicuous consumption than American consumers. These differences between East Asian and Western consumers are considered to be related to Confucianism and a culture of interdependence in East Asia (Wong and Ahuvia 1998). Status concerns had been linked to "education fever" (Sorensen 1994; Kim and Lee 2010) but not yet to fertility.

slope of the fertility-income relationship from positive to negative. This is an interesting result in that Korea displays a positive fertility-income relationship, whereas fertility and income are negatively related in most other countries. We also use our model to revisit the role of social norms on the role of women for fertility choices explored by Myong, Park, and Yi (2021).

We use the quantitative model to study the effect of several government policies. Motivated by actual policies recently introduced in Korea, we consider pro-natal transfers and Pigouvian-style taxes on private investment in education. We find that pro-natal transfers increase the fertility rate as intended, with a magnitude that is in line with recent empirical estimates of the fertility effects of the pro-natal cash bonus in Korea (Kim 2020). Meanwhile, they decrease education spending per child and do so to a greater extent among low-income families. As a result, human capital, output, and consumption are lower in the new steady state. We also investigate the effect of a tax on education spending, as an alternative method of addressing the externality. Such a tax does indeed reduce education spending, while raising fertility. As parents choose to invest less per child, they experience an increase in consumption, while future generations experience lower human capital and consumption. These findings indicate that both policies would necessarily involve some intergenerational conflict by lowering future generations' human capital and output.

Finally, we explore the optimal policy mix that maximizes the welfare of the initial parent generation. We find that the optimal policy is an education investment tax rate of 22 percent and moderately large pro-natal transfers—a monthly child allowance of US\$61 (2.5 percent of average income) for 18 years. This optimal policy increases the fertility rate by 11 percent, lowers the childlessness rate by more than half a percentage point, and decreases education spending by 39 percent. The effects on fertility and education are heterogeneous along the income distribution, with the largest effects felt among the poorest quintile of parents. While this policy addresses the distortion caused by the status externality and is optimal from the perspective of the first generation, we find that it does decrease human capital and output over time. Thus, although the welfare of the parents' generation increases, future generations are worse off compared to a world without the policy, at least in terms of average utility.

We also revisit the policy experiments using a version of our model that incorporates the idea of college slot scarcity with parents who value the college enrollment of their children. We find that the results from this tournament model are similar to those from our status externality model, although education taxes have little effect on fertility and the welfare benefits of the optimal policy mix are somewhat larger. Given these results, we discuss policy implications and avenues for future research.

Our research builds upon the economic analysis of fertility pioneered by Becker (1960) and, more specifically, the quantity-quality trade-off first modeled by Becker and Tomes (1976). Many subsequent analyses have used this framework in quantitative models to understand fertility differences over time and across countries (Greenwood and Seshadri 2002; Greenwood, Seshadri, and Vandenbroucke 2005; Manuelli and Seshadri 2009) and to study the aggregate and distributional implications of fertility in the presence of endogenous human capital investments and heterogeneous households (de la Croix and Doepke 2003; Cordoba, Liu, and Ripoll

2016; Cavalcanti, Kocharkov, and Santos 2020). Some recent work, including that of Sommer (2016); Guner, Kaya, and Sánchez-Marcos (forthcoming); and Daruich and Kozłowski (2020), consider rich heterogeneous agent life cycle models while abstracting from the quantity-quality trade-off. Recently, family policies such as government-subsidized day care and subsidies for having children have been analyzed in quantitative models, though these typically use models with exogenous fertility (Domeij and Klein 2013; Hannusch 2019; Guner, Kaygusuz, and Ventura 2020). See Doepke et al. (2023) for a survey of this literature.

The key novel ingredient in our fertility model is a status externality in education. While a sizable body of work on relative status concerns does exist, dating back to Veblen (1899), the importance of status has not been explored in the context of fertility choices. Though much of the literature analyzes conspicuous consumption, the idea that people may overinvest in education because private returns are greater than social returns has been around for a long time (Akerlof 1976). Ramey and Ramey (2010) argue that competition in college admissions leads to an education externality in the United States, but their analysis also leaves out fertility implications. The macroeconomic implications of status concerns have been analyzed by several authors—e.g., Abel (1990); Gali (1994); and Ljungqvist and Uhlig (2000)—while Genicot and Ray (2017) theoretically explore the effect of status externalities on inequality. None of these papers considers the implications for fertility. The only exceptions are Easterlin (1966), who argues that aspirations formed during childhood affect fertility choices as adults, and Rossi and Xiao (2024), who find sizable spillovers in fertility decisions in China, through various channels, including education.

In contrast to a large part of the fertility literature concerned with excessively high birth rates (e.g., Lee and Miller 1990), we provide a framework in which birth rates can be too low in the presence of a status externality. A few other reasons for fertility being below socially optimal levels have been set forth. Schoonbroodt and Tertilt (2014) argue that the lack of property rights that parents have over their (future) children causes the social benefit from child-bearing to exceed the private benefit, leading parents to choose too few children. Doepke and Kindermann (2019) argue that fertility might be inefficiently low when men and women disagree about fertility and husbands and wives bargain about having children. A similar argument is made in Ashraf, Field, and Lee (2014). Jones (2022) points out that fertility may be too low in a growth model with idea spillovers. Note that these papers use various notions of optimality. Indeed, the definition of efficient levels of endogenous fertility is beset by conceptual and philosophical difficulties. Golosov, Jones, and Tertilt (2007) propose the alternative concept of *A-Efficiency*, which we apply herein.

Finally, there is a largely empirical literature that studies fertility in Korea specifically. For example, Lee (2009) provides a general survey of potential reasons for low fertility and policy responses. The possible connection between the East Asian “education fever” and low fertility was perhaps first pointed out by Anderson and Kohler (2013) in a purely empirical study, also of Korea. Hong et al. (2016) and Kim (2020) both study the recent introduction of pro-natal transfers in Korea and find small but positive effects on fertility. Ma (2016) and Myong, Park, and Yi (2021) set forth an alternative explanation for low fertility in Korea by noting the recent increase in female labor force participation together with strong family values

that expect mothers to stay home with their children.³ We explore social norms on the role of women as an alternative explanation in Section IVB and find it to be less important for our 1970–1975 cohort of women.

The remainder of the paper is organized as follows. In Section I, we describe the data and present our empirical findings on the relationship between fertility and education and spillovers in private education spending in Korea. In Section II, we set up the model. Section III explains how we calibrate the model. Section IV explores the sources of low fertility using our calibrated model. In Section V, we study the effects of pro-natal transfers and education taxes. In Section VI, we solve for the optimal policy mix that maximizes the utility of the first generation. In Section VII, we investigate the tournament model to analyze the idea of college slot scarcity as a source of the externality in education. Section VIII concludes.

I. Fertility, Education, and Education Spillovers

In this section, we show how fertility and private education expenditures in Korea vary along the income distribution. We then provide evidence of externalities across families in private education spending. The documented facts form the basis for the quantitative analysis in this paper.

Our analysis is based on the longitudinal samples of the Korean Labor and Income Panel Study from Korea Labor Institute (2018). We use the cohorts of women born between 1970 and 1975. A robustness analysis in online Appendix A.5 shows similar findings using women born earlier. Fertility is measured by completed fertility.⁴ We will document income gradients in fertility and education investments. As we are interested in a permanent measure of income at the household level, we construct long-term household income as in Chetty et al. (2014) and focus on two-parent households.⁵

A. Fertility and Private Education

Panel A of Figure 1 depicts the relationship between the average number of children and average household income in each income quintile. The relationship is generally positive: the number of children increases from 1.80 for the lowest-income quintile to 2.03 in the highest-income quintile. This finding suggests that low birth rates in Korea are related to factors affecting low-income households. One way to quantify the relationship between fertility and income is to estimate the income

³ A family values explanation has similarly been given for certain low-fertility European countries (Billari and Kohler 2004). However, these countries have comparatively higher birth rates than Korea, and in some cases, fertility has recently been trending upward again. Thus, Anderson and Kohler (2013) argue that strong social norms alone cannot fully explain the Korean case.

⁴ Completed fertility is the average number of children ever born, typically measured after age 40. It is 1.9 in our data for the 1970–1975 cohort, which is higher than the total fertility rate (TFR) of 0.9 in 2019. Naturally, TFR is lower than completed fertility, reflecting the declining fertility trend in Korea.

⁵ We use the average income of two-adult households in which the woman's age is between 40 and 43. Focusing on two-adult families enables us to abstract from the different fertility choices of single versus married women and to circumvent comparisons between the income of one- and two-adult households. However, as the fraction of one-adult households is relatively small (8 percent) in our data, including single women does not significantly change our results, as shown in online Appendix A.6.

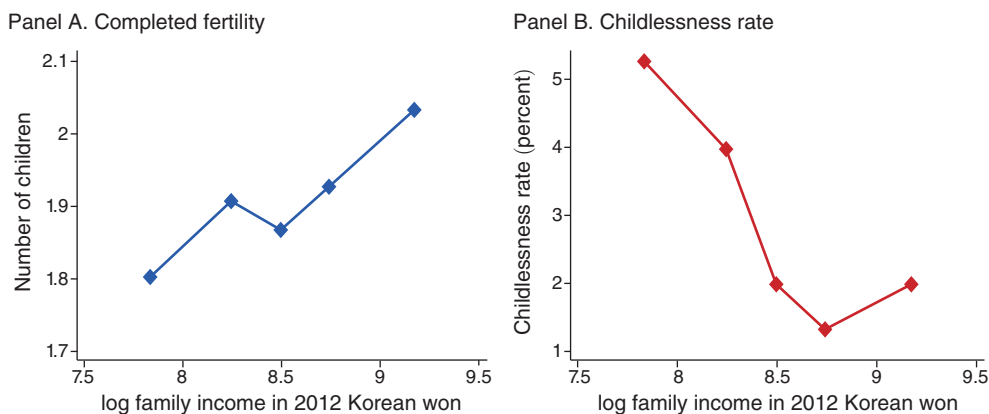


FIGURE 1. FERTILITY BY INCOME QUINTILE

Note: We group households into quintiles based on their long-term income and calculate average completed fertility and the childlessness rate in each quintile for cohorts born in 1970–1975.

elasticity of demand for children following Jones and Tertilt (2008). It is 0.082 in our data.

The positive income-fertility relationship may appear puzzling given that fertility and income are negatively related in most countries and over time. For example, Jones and Tertilt (2008) find negative elasticities for all cohorts of US women going back to 1825.⁶ Elasticity estimates from the US cohorts that most closely correspond to our data are around -0.2 .

We also consider childlessness, the extensive margin of fertility choice. Panel B of Figure 1 shows that the childlessness rate has a negative relationship with family income, decreasing from 5.3 percent for the lowest-income quintile to 2 percent in the highest-income quintile. Poor families thus are substantially more likely to have no children. Although the absolute level of childlessness is low in Korea, its clear income gradient confirms that the extensive margin is also important to understanding the positive relationship between fertility and family income.

Next, consider private education. Korean children spend much time in after-school programs and with private tutors. While some of these activities are related to physical education, art, and music (and may be considered leisure activities), a large portion of children also attend after-school programs to learn English and improve their math and computer skills (see Table 1). The high participation rate is even more striking given the fact that private education is expensive in Korea. In our data, we find that total education expenditures per child amount to 9.2 percent of family income.⁷ Thus, education spending is surprisingly high in Korea even though

⁶Jones and Tertilt's (2008) estimates are based on men's income. When male income is used instead of family income in our dataset, the relationship between fertility and income remains positive, with an estimated elasticity of 0.096. An exception to the negative fertility-income relationship in the United States is the most recent cohorts. Bar et al. (2018) document that fertility is upward sloping for very high incomes. While they relate this phenomenon to the marketization of time, concerns about relative education might also play a role.

⁷Specifically, we begin by calculating the education expenditures and incomes separately by child's age from birth to age 24, which corresponds to the age range we use in our model. We then divide the sum of expenditures by the sum of after-tax income. See online Appendix A.2 for details.

TABLE 1—PRIVATE EDUCATION PARTICIPATION RATE, 2019 (PERCENT)

	Average	Elementary	Middle	High
Any subject	74.8	83.5	71.4	67.9
A. Main subjects	56.7	57.9	61.8	57.8
a. Individual tutoring	8.9	6.4	10.4	14.3
b. Group tutoring	9.3	10.5	10.0	7.6
c. <i>Hagwon</i>	41.1	37.6	50.7	46.0
d. Others ^a	20.4	32.1	11.3	6.9
B. Art, music, physical activities	44.0	67.4	26.2	15.3
a. Individual tutoring	5.6	7.3	4.6	3.6
b. Group tutoring	5.0	7.8	3.1	1.0
c. <i>Hagwon</i>	34.8	55.9	17.1	10.3
d. Others ^a	5.8	9.2	3.4	1.3

Note: The main subjects include Korean, English, math, science, second foreign language, writing, and computer science.

^aOthers may include online education programs or home-based sessions with tutors from education companies. Note that the total is not the sum of the lower categories since a single child may, for example, attend a *hagwon* and have individual tutoring lessons.

Source: Statistics Korea (2019)

empirical evidence shows that returns to private education are quite low by international comparison (Psacharopoulos and Patrinos 2018).

To assess how education expenditures vary across the income distribution, we use detailed information on private education expenditures, disaggregated by child's age, from survey data. Table 2 displays the percentage of family income spent on private education by quintile for children at different stages of education. Poorer families spend a higher percentage of income on private education at most stages. The last column, which shows overall education spending over a child's entire education, reveals that parents in the bottom quintile spend 3.3 percentage points more of their family income than those in the top quintile. The income elasticity of private education is equal to 0.698, which is substantially less than 1.

B. Spillovers in Private Education Spending

We now provide empirical evidence for spillovers in education investment across families. We do this by analyzing how parental investment decisions are affected by others in the same region. The presence of such spillover effects motivates our model of status externalities in education developed in the following section. Specifically, we estimate the effects of spending in private education among relatively rich families on education spending of other families. To address potential endogeneity issues, we exploit recent changes of curfews that banned private education beyond a certain time of the day (10 PM or 11 PM) implemented across regions in different years (Choi and Choi 2016; de Silva 2021). The curfews reduced participation in private education programs, and hence spending, largely among the rich.⁸ We thus

⁸Curfews do not affect most elementary school students. We focus on parents with middle school students in 2009–2015 because actual curfew variations are most prominent; see online Appendix A.3 for details. We do not estimate the effects on fertility since most parents of middle school children have completed their fertility decisions and cannot easily adjust along this margin.

TABLE 2—PERCENTAGE OF INCOME SPENT ON PRIVATE EDUCATION

Income quintile	Preschool	Elementary school	Middle school	High school	Weighted average
1st	9.0	9.0	8.4	5.7	8.4
2nd	6.9	8.0	8.5	6.1	7.4
3rd	6.1	7.7	7.6	6.6	7.0
4th	5.6	6.7	7.4	6.9	6.5
5th	4.6	5.0	5.8	5.8	5.1

Notes: This table shows the percentage of income spent on private education per child at each stage of education. The shares are calculated by dividing the average expenditures by the average income of the corresponding income quintile. The last column shows the average expenditures across different education stages weighted by the number of years spent in each education stage. See online Appendix A.2 for more details. The online Appendix also includes the level of spending and the fraction of households with no private education spending by income quintiles (see Tables A2 and A3).

use the curfews as instrumental variables (IV) for education spending among the rich.

To satisfy the exclusion restriction, it is important that the dependent variable is not directly affected by the curfew changes. In other words, ideally one should consider only families whose children were not attending *hagwons* after 10 PM. Although this information is not available in our baseline data, data from the Korean Time Use Survey in 2009 (Statistics Korea 2009) reveal that only around 15 percent of students attend *hagwons* at 10 PM or later and that families with low socioeconomic status are much less likely to send their children to a *hagwon* late at night, as shown in online Appendix Figure A4. As such, the dependent variable of individual private education spending is constructed based on families with income below the median (*low income*) or those where both parents have at most a high school degree (*low education*).

Formally, we estimate the following two-stage least squares regression:

$$(1) \quad \ln E_{st} = \delta_{10} \Lambda_{st}^{10} + \delta_{11} \Lambda_{st}^{11} + \gamma' \mathbf{X}_{ist} + \epsilon_{ist},$$

$$(2) \quad \vartheta_{ist} = \beta \ln \hat{E}_{st} + \eta' \mathbf{X}_{ist} + \varepsilon_{ist}.$$

In the first stage, we regress the log of E_{st} , private education spending per student among the rich in region s at time t , on a set of indicator variables for the presence of a curfew at 10 PM and 11 PM, Λ_{st}^{10} and Λ_{st}^{11} , respectively.⁹ Then, in the second stage, we regress ϑ_{ist} , private education spending per student relative to overall household expenditure of (either low-income or low-education) family i in region s at time t , on the log of the predicted regional spending $\hat{E}_{st}$ of high-income families.¹⁰ This gives the main coefficient of interest β , which captures the spillover effect.

⁹Since 12 PM curfews are in place for all provinces during the estimation periods (as shown in online Appendix Figure A3), the effects of the 10 PM and 11 PM curfews are relative to the 12 PM curfew.

¹⁰Regional averages based on our KLIPS samples can be noisy since the number of samples becomes very small at some regions. Therefore, we rely on the Private Education Expenditures Survey microdata (Statistics Korea 2013, 2018) to compute E_{st} . See online Appendix A.3 for details.

TABLE 3—ESTIMATION OF PRIVATE EDUCATION SPILLOVERS USING INSTRUMENTAL VARIABLES

Samples:	Low income			Low education		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. 2nd stage</i>						
β	0.065	0.091	0.049	0.039	0.070	0.042
SE	(0.014)	(0.030)	(0.038)	(0.015)	(0.039)	(0.043)
<i>Panel B. 1st stage</i>						
δ_{10}	-0.146	-0.110	-0.092	-0.148	-0.106	-0.089
SE	(0.026)	(0.029)	(0.031)	(0.024)	(0.025)	(0.028)
δ_{11}	-0.060	-0.031	-0.033	-0.054	-0.019	-0.019
SE	(0.039)	(0.037)	(0.035)	(0.035)	(0.033)	(0.031)
<i>F</i> -stat	16.6	7.5	4.4	20.0	9.2	5.2
Trend ^a	No	Yes	No	No	Yes	No
Year FE	No	No	Yes	No	No	Yes
Observations		1,501			1,528	

Notes: Each column presents the estimation results from (1) and (2). Panel A shows the main coefficient estimates from the second-stage regression. Panel B shows the first-stage regression results. We report the wild bootstrap clustered standard errors (Cameron, Gelbach, and Miller 2008) in parentheses, given the relatively small number of clusters.

^a“Trend” means that the regression includes a linear time trend.

To compute the regional spending among the rich (E_{st}), our baseline estimation uses the top 15 percent income group to maximize the validity of our instruments and to minimize potential biases from defining the rich too narrowly or too broadly, as detailed in online Appendix A.3. Both equations include a vector of control variables in $\mathbf{X}_{ist}$, such as family income, parental education, parents’ age groups, and province fixed effects. Since curfew changes over time tend to be synchronized across provinces, as shown in online Appendix A.3, curfew indicators are correlated with time fixed effects, which would adversely affect the efficiency of our estimation. Thus, we consider both specifications, with and without year fixed effects.

Panel A of Table 3 shows that the estimated β is positive and statistically significant for both samples considered in the specifications without year fixed effects (columns 1 and 4). The estimated β remains significant when we add a linear time trend to the regression. When adding time fixed effects instead, not surprisingly, the instruments are weaker and the estimates become more noisy and even insignificant. Thus, while the coefficient remains of the same order of magnitude across all specifications, it is somewhat imprecisely estimated. To interpret the size of the estimates, a value of $\beta = 0.05$ implies that a 10 percent reduction in the top 15 percent income households’ spending on private education decreases spending for the lower half of the income distribution by around 0.5 percentage points.¹¹

Panel B of Table 3 shows that the first-stage *F*-statistic is large in all cases, confirming the relevance of our instruments. Notably, the estimated values of δ_{10} and δ_{11} are negative, and δ_{10} is much larger in absolute value than δ_{11} , with smaller standard errors. This implies that the earlier curfews reduced regional private education spending per child by more than the later curfews. This is reasonable because a later

¹¹ The average ϑ_{ist} is around 6 percent in the sample used for the estimation.

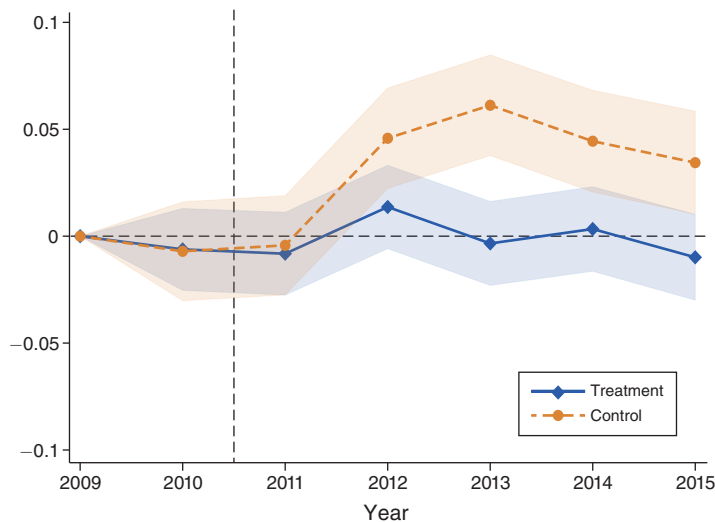


FIGURE 2. CURFEW CHANGES AND TRENDS IN PRIVATE EDUCATION SPENDING

Notes: This figure shows the average log private education expenditure per child for the treatment (with a curfew change since 2011) and control groups (with no curfew change). The lines are normalized to 2009 after controlling for year and province fixed effects. The shaded areas indicate the 95 percent confidence intervals. See online Appendix A.3 for details.

curfew is less restrictive by definition and also because only a few students (around 3 percent, as shown in online Appendix Figure A4) attended a *hagwon* after 11 PM.

To inspect if curfew adoptions across provinces were plausibly exogenous, we compare the log of private education spending for two groups of provinces using the Private Education Expenditures Survey microdata in Figure 2. The first group (treatment group) includes families in provinces with curfew changes from 2011 to 2013, and the other group (control group) includes families in provinces with no changes since 2009.¹² Clearly, pre-trends are not distinguishable between the two groups, suggesting that the curfew adoption variations are unlikely due to policy reactions in response to a rising pre-trend in regional private education spending. Figure 2 also shows that the private education expenditure levels are lower in the treated regions in the posttreatment periods, in line with the negative coefficients of the curfew dummies in the first stage of the IV regression (panel B of Table 3).

We also explore the trends of private education spending for different income groups. We find that the differences between the groups become statistically most pronounced when using the top 15 percent income families (instead of the top 50 percent), reinforcing our choice of the top income group in the first stage and in the structural model presented in the next section. See online Appendix Figure A5 for details.

¹²Figure 2 excludes one province, Daejeon, where the curfew changed in 2009, the start year of the analysis window. Including Daejeon does not change things considerably.

II. The Model

Our model builds on the quantity-quality literature where parents choose the number of children and how much to invest in each. The model is an overlapping generations model, where each generation lives for two periods: as children and as adults. For expositional convenience, we present the model in a recursive formulation and let a variable with a prime denote its value in the next period. We abstract from marriage and assume asexual reproduction. Similarly, we do not distinguish between boys and girls and thus abstract from gender differences in parental inputs.¹³ We assume that fertility is discrete. This is not only realistic but also naturally leads to childlessness as an equilibrium outcome.¹⁴ For some parents, the cost of having a child will be so high (relative to income) that they may prefer not to have any offspring. Parents also choose education investments into their children, which, together with luck, determines the human capital of the children once they are adults.

As documented in Section I, fertility and private education investment vary across the income distribution. To reproduce these empirical patterns in the model, we introduce several dimensions of heterogeneity. Potential parents differ in their own levels of human capital, and hence income, which leads them to make different fertility and education investment choices for their children. However, since human capital is endogenous in the model (each parent is a child of a previous parent), we introduce two exogenous sources of heterogeneity so that the income distribution in equilibrium is nondegenerate. First, parents differ in their relative taste for own consumption relative to leisure and utility from offspring. Second, parents draw their child's idiosyncratic ability that is correlated across generations, which enters into the human capital production function, as detailed below.¹⁵

The novel feature of our model is a *status externality*. There are two standard ways to model such externalities in the literature. Abel's (1990) *catching up* with Joneses assumes that utility depends on the *previous* period's consumption, whereas Gali's (1994) *keeping up* with the Joneses considers utility that depends on *current* aggregate consumption. In our model, parents care about the quality of their children relative to other children. Specifically, we model this by assuming the following functional form in the utility parents derive from the number n and the human capital (or wage in adulthood), h' , of their children:

$$(3) \quad \phi(n) \log(h' - \chi \tilde{h}'),$$

¹³ In reality, parental inputs are higher for boys than girls. But the gaps have substantially declined in Korea over time, so that current gender gaps are small (Choi and Hwang 2020).

¹⁴ The importance of distinguishing the extensive and intensive fertility margins was recently emphasized by Aaronson, Lange, and Mazumder (2014) and Baudin, de la Croix, and Gobbi (2015). We do not consider infant mortality in our model, as it is negligible in South Korea.

¹⁵ Having both sources of heterogeneity gives significant flexibility to the shape of the fertility-income relationship and allows for both directions of causality. Parents with higher human capital may choose to have more children because of an income effect or fewer because of a substitution effect. They may also react to the quantity-quality trade-off differently. Finally, potential parents with strong preferences for leisure work less and hence have less income. This last channel is less common in the literature but is considered in Jones, Schoonbroodt, and Tertilt (2010).

where $\tilde{h}'$ is the benchmark human capital, which is a statistic from the marginal distribution of children's human capital $\mathfrak{F}(h')$.¹⁶ The strength of the externality is determined by $\chi \geq 0$. The status externality may in fact originate from deeper sources, such as intense aspirations (Genicot and Ray 2017) or the distinct school grading system in Korea that heavily uses norm-referenced evaluations. As the common theme of these various sources is parents' concern about their children's relative performance, we assume that the benchmark is children's human capital.

Parents also derive utility from the number of children, captured by $\phi(n)$. We require this function to be increasing and concave. When parents choose the number of children, n , in addition to investing into their human capital, they also have to pay a fixed time cost of λ per child.

We assume that children's human capital is formed through three different inputs: an idiosyncratic (genetic) ability component κ that is stochastically inherited from parents, baseline human capital θ , and private education investments x . Ability is correlated across generations following a Markov chain, drawn from a discrete distribution: $\kappa \in \{\kappa_k\}_{k=1}^{N_\kappa}$ with the transition probability of π_{jk}^κ , where j denotes the ability index of a parent with ability κ_j^P . This seeks to capture the fact that high-ability parents are more likely to have children with high abilities.¹⁷ Parents decide on the amount x to invest in each child after κ is realized. We assume that these are complementary and choose a functional form similar to the specifications used in de la Croix and Doepke (2003) and Cavalcanti, Kocharkov, and Santos (2020):

$$(4) \quad h' = A_h \kappa (\theta + x^\alpha),$$

where $\alpha \in (0, 1)$. The parameter $\theta > 0$ guarantees that even if parents choose zero education for their children, the children will have some baseline human capital. This can capture raw intelligence and publicly provided education, for example. $A_h > 0$ adjusts the scale of the human capital production.

A parent's ex post preferences (after all uncertainty is resolved) can then be summarized by the following utility function:

$$(5) \quad U(c, l, n, h', \tilde{h}') = b \log\left(\frac{c}{\Lambda(n)}\right) + \nu \frac{l^{1-\gamma}}{1-\gamma} + \phi(n) \log(h' - \chi \tilde{h}'),$$

where b is the preference type mentioned above. Some people intrinsically prefer a "market-consumption lifestyle" (i.e., high b), whereas others intrinsically prefer a "family-leisure lifestyle" (i.e., low b). We assume that b can take N_b possible values: $b \in \{b_i\}_{i=1}^{N_b}$ with the corresponding probabilities $\{\pi_i^b\}_{i=1}^{N_b}$. As is standard in the literature, utility depends on household consumption expenditures c divided by the household equivalence scale, denoted by $\Lambda(n)$. Parents value leisure l , weighted by $\nu > 0$. The curvature of utility from leisure is captured by $\gamma > 0$. Unlike b , both ν and γ are assumed to be identical across people.

¹⁶We adopt the same functional form of externality as in Ljungqvist and Uhlig (2000). As the problem is recursive, children will face the same externality when they become parents.

¹⁷Thus, in our model children's earnings are related to their parents' earnings for two reasons: genetics and investments. This dual role of parents is supported by empirical evidence, e.g., in Fagereng, Mogstad, and Rønning (2021), who compare adopted and own children.

The timing within a period is as follows. Parents start the period endowed with human capital h and their own ability κ^p (which was relevant during their childhood). The preference type b is then realized. Given b , agents choose fertility n , taking into account expectations about their potential child's ability. Once the children's ability κ is realized, parents make decisions regarding education investments, labor supply, and consumption.

Given the above timing assumptions, the parental decision problem can be summarized in two steps. First, the parent makes a discrete fertility choice, not yet knowing the ability of her children. For simplicity, we assume that all children of a given parent have the same ability. The choice problem in the first step is thus

$$(6) \quad \max_{n=0,1,\dots,N_n} \left\{ \mathbb{E}_{\kappa|\kappa^p} V(h, b, \kappa, n; \tilde{h}) \right\},$$

where $V(h, b, \kappa, n; \tilde{h})$ is the household's value of having n children of type κ . The second step optimization gives a value function defined as

$$(7) \quad V(h, b, \kappa, n; \tilde{h}) = \max_{c,x,l} \left\{ b \log\left(\frac{c}{\Lambda(n)}\right) + \nu \frac{l^{1-\gamma}}{1-\gamma} + \phi(n) \log(h' - \chi \tilde{h}') \right\}$$

subject to

$$\begin{aligned} c + xn &\leq wh(1 - \lambda n - l), \\ h' &= A_h \kappa(\theta + x^\alpha), \\ l &\in [0, 1 - \lambda n], \quad \tilde{h}' = \Gamma(\tilde{h}). \end{aligned}$$

This optimization problem shows that parents invest in their children's education given the number of children and the realized ability shock. The household's time endowment is normalized to one. The total time available for labor supply and leisure decreases with the number of children since each child requires λ units of time. Denoting leisure as l , labor supply in efficiency units is given by $(1 - \lambda n - l)h$. Finally, the Γ function describes the law of motion for $\tilde{h}$ as perceived by households. In equilibrium, this should be consistent with the actual law of motion for $\tilde{h}$. Note that parents care about children in a warm-glow fashion, as parents are not altruistic in the Barro-Becker sense (Becker and Barro 1988; Barro and Becker 1989).¹⁸

Children are costly in our model for several reasons. First, each child requires an education investment in the form of the endogenous expenditure x . Second, a larger household with more children would reduce utility through the household equivalence scale channel $\Lambda(n)$ since the same consumption expenditure must be shared by more people in the household. Finally, children require time inputs, which reduces the time that can be used for leisure or labor supply. Note that although parents

¹⁸ A fully altruistic model with status externalities would lead to a complicated game across generations and dynasties, as parents would need to consider their own children's choices as well as those of other dynasties when choosing education inputs. This is different in the standard Barro-Becker model, which collapses to a planning problem. With externalities, the equivalence between the equilibrium and a planning problem breaks down; see Golosov, Jones, and Tertilt (2007) for details.

would always have a positive expected utility when they choose to have a nonzero number of children (otherwise, they would prefer childlessness), we do not rule out the possibility of negative ex post utility from children. This is possible when the children's ultimate level of human capital is low (specifically, $h' - \chi \tilde{h}' < 1$ in our case), which would be the case among unlucky parents of modest means whose children's idiosyncratic ability is low.

Finally, to close the model, we assume that aggregate production is linear in aggregate labor. Specifically, letting L be average efficiency units of labor, output per capita is given by $Y = AL$, where A is total factor productivity.

The equilibrium definition is relatively standard and given in online Appendix B.1. Given the status externality, the key object of the equilibrium both in the steady state and along the transition path is the endogenous distribution of human capital. This distribution is determined as a fixed point that agents take as given, while their expectations must be consistent with the actual evolution of the human capital distribution. Further details on how we compute the steady-state equilibrium and equilibrium transitional dynamics are provided in online Appendix B.1.

III. Calibration

We solve the model numerically, as detailed in online Appendix B.1, and calibrate it to the KLIPS samples described in Section I. The calibration proceeds in two steps. First, some parameter values are chosen externally based on direct data analogs, the literature, or simple normalization. Second, the remaining parameters are chosen to match relevant data moments. We also examine the model fit.

We start with some preliminaries. We set the maximum number of children equal to three, $N_n = 3$, because the portion of households with more than three offspring is very small.¹⁹ Because fertility is a discrete choice between zero and three, it is unnecessary to impose a parametric functional form on $\phi(n)$. Instead, we let it be nonparametric and assume that $\phi(n) = \phi_n$ and $\phi(0) = 0$, leading to three parameters: $\{\phi_n\}_{n=1}^3$.

We let the discrete distribution for b approximate a log-normal distribution:

$$(8) \quad \log b \sim \mathcal{N}(0, \sigma_b^2),$$

with $N_b = 20$. We let the Markov chain for κ approximate an AR(1) process:

$$(9) \quad \log \kappa = \rho_\kappa \log \kappa^P + \varepsilon_\kappa,$$

where $\varepsilon_\kappa \sim \mathcal{N}(0, \sigma_\kappa^2)$, using Tauchen (1986) with $N_\kappa = 20$. This leads to two parameters— ρ_κ and σ_κ —to be calibrated for the Markov chain. Our quantitative results are barely affected by N_κ as long as it is sufficiently large.

¹⁹We include the small fraction of women with four and five children (1.59 percent in the data) in the three children category. See online Appendix Table A1 for details.

A. Externally Calibrated Parameters

The parameter γ governs the curvature of the utility function with respect to leisure. We choose $\gamma = 2$ so that the implied intertemporal elasticity of substitution is 0.5, as is standard in the literature. The parameter λ captures the time costs of children. In our samples from the KLIPS, the average time parents of children under 18 spend with each child is 5.7 hours per week, as detailed in online Appendix A.7. Assuming a total weekly time endowment of 100 hours, this leads to $\lambda = 0.041$.²⁰ We consider this time input to be constant. In contrast to the United States, where more educated parents spend more time with their children (Guryan, Hurst, and Kearney 2008), we find no meaningful educational gradient in our data, as reported in online Appendix Table A8.

For the household equivalence scale $\Lambda(n)$, we use the OECD modified equivalence scale, which assigns 1 to the adult head, 0.5 to an additional adult, and 0.3 to each child. Finally, the TFP parameter A is normalized to one.

B. Internally Calibrated Parameters and Model Fit

The remaining 11 parameters are $\phi_1, \phi_2, \phi_3, \sigma_\kappa, \nu, \sigma_b, \chi, \theta, \alpha, \rho_\kappa$, and A_h . We calibrate them to match 12 moments from the data described below, by minimizing the sum of squared percentage differences between data and model moments. All data moments are either directly taken from Section I or described in online Appendix A.

The data moments include the fraction of families with one, two, and three or more children. We further include the Gini coefficient of our long-term income as a measure of inequality. To capture the relationship between income and fertility, we include the income elasticity of fertility as well as the fraction of childless families in the poorest quintile. Since spending on education is a key object of interest in our model, we include both average education spending relative to income and the income elasticity of education spending. We also include a measure of the education spillover based on our empirical estimates in Section IB. To connect to the reduced-form estimate, we define the benchmark human capital to be the conditional mean among the top 15 percent of the human capital distribution.²¹ Finally, intergenerational income elasticity is included as a measure of income persistence across generations. Table 4 reports all data moments as well as the model analogs and the calibrated parameter values.

Since every moment that results from the model is a function of all parameters, there is no one-to-one link between parameters and moments. However, some moments are more informative for particular parameters. It is thus instructive to explain these intuitive links, even though there is no formal identification procedure. The first three parameters in Table 4 govern the utility of having n children. These utility parameters are directly related to the percentages of people with zero, one, two, and three children in the data. The calibrated values of these parameters are

²⁰ A disposable time endowment of 100 hours per week is fairly standard in the literature, as it leaves 68 hours for sleep and personal care. We assume children are costly for parents until they become adults, i.e., for 18 years. Since a model period is 25 years, we further adjust the time cost by a factor of 18/25, which then gives $\lambda = 0.041$.

²¹ The benchmark being high socioeconomic status is also in line with the empirical finding that comparison motives are largely upward looking (Ferrer-i-Carbonell 2005).

TABLE 4—INTERNALLY CALIBRATED PARAMETERS AND MOMENTS: MODEL VERSUS DATA

Parameter and interpretation		Moment	Model	Data
$\phi_1 = 1.39$	Utility from number of children	Pr(# child = 1)	0.176	0.196
$\phi_2 = 2.18$		Pr(# child = 2)	0.633	0.631
$\phi_3 = 2.57$		Pr(# child ≥ 3)	0.158	0.144
$\sigma_\kappa = 0.361$	Ability dispersion	Gini income	0.266	0.263
$\nu = 1.87$	Leisure constant	Avg total hours worked	0.290	0.301
$\sigma_b = 0.587$	Preference dispersion	Income elasticity of fertility	0.080	0.082
$\chi = 0.078$	Status externality	Education inv. spillover estimate	0.038	0.039
		Childless in first income quintile	0.052	0.053
$\theta = 1.08$	HK production technology	Avg. investment-income ratio	0.110	0.092
$\alpha = 0.668$		Income elasticity of investment	0.775	0.698
$\rho_\kappa = 0.376$	Ability persistence	Intergenerat. income elasticity	0.327	0.320
$A_h = 2.23$	HK scale	Output per capita (normalization)	0.958	1.00

Note: All data moments are based on KLIPS data as described in Section I and online Appendix A.

$\phi_1 = 1.39$, $\phi_2 = 2.18$, and $\phi_3 = 2.57$, which are increasing at a decreasing rate with n .

The parameter σ_κ determines the variability of the idiosyncratic ability component in child human capital development. Because this ability shock is an important exogenous source of income heterogeneity in the model, it is largely determined by the Gini coefficient in the data (0.263). Note that our Gini coefficient is somewhat lower than what is typically reported (e.g., by the OECD) because ours is based on income averaged over several years rather than annual income, as described in online Appendix A.1. The calibrated value of σ_κ is 0.361.

The value parents place on leisure relative to consumption and children is given by the parameter ν . It is pinned down by the average total hours worked in the data (30.1 hours per week or 0.301), as described in online Appendix A.7. This value is based on both members of the household and includes both intensive and extensive margins. This leads to a calibrated value of $\nu = 1.87$.

Next, σ_b controls the degree of heterogeneity in preferences. An important reason for having preference heterogeneity is to allow for flexibility in shaping the model-generated relationship between income and fertility. Specifically, a greater variability of b makes the equilibrium fertility-income relationship less positive (or more negative). Our calibrated value of $\sigma_b = 0.587$ allows the model to match the positive income elasticity of fertility of 0.082 from the data almost perfectly.

The strength of the status externality is governed by χ . To discipline this parameter in our structural model, we use our reduced-form estimate in Section IB in accordance with Nakamura and Steinsson (2018), who highlight the approach of using reduced-form causal estimates for the calibration of structural macro models. Given the uncertainty in the point estimates of spillover effects in Table 3, we choose a conservative value of 0.039 as a target moment. Online Appendix B.2 explains how we compute the model counterpart of the spillover effect. The externality is also relevant for childlessness. Intuitively, a strong status concern leads parents to spend much on education, making children costly. For some parents, especially at the bottom of the income distribution, children become so costly that they prefer to have none. Thus, we include the childlessness rate for the bottom quintile as an additional

TABLE 5—FERTILITY AND EDUCATION SPENDING ACROSS INCOME QUINTILES

		Income quintile					
		All	1st	2nd	3rd	4th	5th
Completed fertility	Data	1.91	1.80	1.91	1.87	1.93	2.03
	Model	1.92	1.78	1.89	1.94	1.99	1.97
Childlessness rate (percent)	Data	2.9	5.3	4.0	2.0	1.3	2.0
	Model	3.4	5.2	2.9	3.1	3.0	2.7
Education spending per child rel. to income (percent)	Data	9.2	11.2	9.9	9.3	8.7	6.9
	Model	11.0	13.6	12.1	11.2	10.5	10.2

Notes: All data moments are based on KLIPS data as described in Section I and online Appendix A. Education spending is higher than reported in the last column of Table 2. This is because the income gradient of education expenditure in Table 2 is estimated using individual-level survey data that contain only a subset of total household expenditures on education (e.g., excluding out-of-pocket school tuition); we thus scale the data to align with the mean of a more comprehensive measure of total education expenditures (9.2 percent of income) based on household-level survey data.

target moment. The calibrated value of $\chi = 0.078$ allows the model to match both the spillover estimate and the childlessness rate of 5.3 percent quite successfully.

The human capital production function includes two parameters: θ and α . Since θ decreases the marginal return of education spending, it reduces the incentive to invest in education and is therefore useful to match average education spending. Average education spending per child relative to income is 9.2 percent in the data. The calibrated value of θ is 1.08. In the data, the income elasticity of private education spending based on quintile averages is 0.698. This moment is useful for pinning down α because it shapes the marginal product of education investments. Our calibrated α is 0.668.

Note that ρ_κ governs the strength of the exogenous ability transmission from parents to children. This strongly affects how income is correlated across generations. Therefore, our last target statistic is the intergenerational elasticity of income, which is 0.32 in the data. This is the mean value of the estimates based on different age combinations from two generations, as reported in online Appendix A.4, and is close to its US counterpart, 0.341 (Chetty et al. 2014). The calibrated value of ρ_κ is 0.376.²² Finally, the scale parameter A_h is calibrated to be 2.23 to normalize the steady-state output per capita of one.

Table 5 shows that the model does a good job of matching the fertility rate across income quintiles in terms of both completed fertility and childlessness rates. Specifically, the model reproduces the average number of children born to parents in the bottom income quintile (around 1.8), as well as the higher average number of children in the richest income quintile, at 2.0. Because we had included the fraction of families with a given number of children as targets, the model unsurprisingly matches the average fertility rate of 1.91 quite well. Moreover, both in the model and the data, the childlessness rates in the third to fifth income quintiles are quite flat, whereas they are higher in the first income quintile. Overall, the average childlessness rates are around 3 percent both in the model and in the data.

²²If instead we set $\rho_\kappa = 0$ and keep all other parameters as in the baseline calibration, the intergenerational income elasticity would be only 0.104.

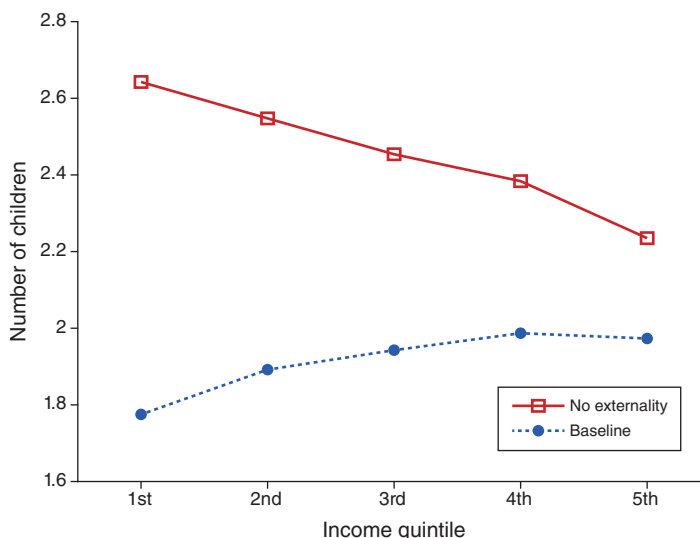


FIGURE 3. STATUS EXTERNALITY AND FERTILITY ACROSS INCOME QUINTILES

Table 5 also reports education expenditures per child relative to income in the model and in the data. Since we target the overall slope (i.e., the income elasticity of investment), the model successfully generates the decreasing pattern of private education expenditures across income quintiles, as observed in the data. In the model, parents in the top income quintile spend 10 percent of their income on education per child, whereas those in the bottom quintile spend a substantially larger fraction of their income on education (nearly 14 percent of total income per child).

IV. Sources of Low Fertility

In this section we explore the role the status externality plays for low fertility. We compare it to the alternative hypothesis of social norms on the role of women.

A. Status Externalities in Education

To understand the role of the status externality for fertility choices, we turn it off by setting $\chi = 0$. Figure 3 shows how fertility rates across income quintiles change in the absence of the externality. Two points are worth noting. First, the fertility rate increases across all income quintiles. The aggregate fertility rate, at 2.45 births per woman, is considerably higher than the value of 1.92 in the baseline model with the status externality. Second, the increase in fertility rates is relatively higher among low-income groups. The income elasticity of fertility takes a negative value in the model without the externality (-0.114), starkly contrasting with the elasticity of 0.080 in the baseline model. This finding is interesting since the positive relationship between income and fertility disappears when the status externality is removed, bringing it more in line with countries such as the United States (Jones and Tertilt 2008). This result suggests that the status externality may be an important factor behind the positive income elasticity of fertility documented in Section I.

TABLE 6—EFFECTS OF THE STATUS EXTERNALITY ACROSS INCOME QUINTILES

		Income quintile				
		1st	2nd	3rd	4th	5th
Childlessness rate (percent)	Baseline	5.2	2.9	3.1	3.0	2.7
	No externality	0.2	0.3	0.5	0.8	1.2
Education spending per child rel. to income (percent)	Baseline	13.6	12.1	11.2	10.5	10.2
	No externality	5.0	5.1	5.8	5.7	6.3
Change relative to baseline (percent)		−63.0	−57.8	−48.4	−45.8	−37.6

The top panel of Table 6 also shows that childlessness rates are generally much lower—and increase with income—in the absence of the externality. This is in sharp contrast to the baseline model, which features higher childlessness rates, especially among the lowest-income households (around 5 percent). This indicates that the status externality puts more pressure on low-income households in our baseline model since childlessness falls substantially in the first income quintile when the externality is removed.

The bottom panel of Table 6 shows what parental expenditures on private education relative to income would be in an economy without the externality. First, households would generally spend less on private education in the absence of a motivation to catch up with the children of others. This effect is relatively stronger among low-income households, reducing investment per child relative to income by 63 percent. The reason is that low-income parents in our baseline model invest more aggressively (measured by the percentage of family income spent on private education), in the hope of bringing their children's status closer to that of the other children.²³

B. Social Norms about the Role of Women

Myong, Park, and Yi (2021) propose an alternative hypothesis for the low fertility rates in South Korea related to social norms about the role of women. They argue that a social norm about the unequal gender division of childcare coupled with an increase in female wages making such a gendered division more costly is the main reason behind the low fertility rate. We now explore this hypothesis in our model. While we have no distinction between mothers and fathers in our model and hence cannot speak to the gendered division directly, we can explore this idea as follows. The reason for the social norm to depress fertility is that for some couples (those where the wife has a higher wage than the husband) it would be efficient for the fathers to provide all the childcare, yet, if due to the social norm the mother still does most of it, this comes at a financial cost in terms of lost family income. The higher this lost income, the more expensive are children, which lowers demand for children. Myong, Park, and Yi (2021) estimate the cost of the social norm to be

²³ This effect echoes the empirical findings of Bertrand and Morse (2016), who find that poor people spend a larger share of income on housing, especially when they live close to high earners, which the authors interpret as status-seeking behavior.

TABLE 7—EFFECTS OF REMOVING SOCIAL NORMS ON FERTILITY AND EDUCATION

		All	Income quintile				
			1st	2nd	3rd	4th	5th
Completed fertility	Baseline	1.92	1.78	1.89	1.94	1.99	1.97
	No social norm	1.97	1.86	2.00	1.97	2.01	2.00
Childlessness rate (percent)	Baseline	3.4	5.2	2.9	3.1	3.0	2.7
	No social norm	3.2	4.7	3.0	2.9	2.7	2.8
Education spending per child rel. to income (percent)	Baseline	11.0	13.6	12.1	11.2	10.5	10.2
	No social norm	10.7	12.9	11.4	10.8	10.4	9.9

Note: The removal of social norm is captured by a reduction in time costs of children (λ) by 4.3 percent, the estimate according to Myong, Park, and Yi (2021).

4.3 percent of the average time cost of raising a child. Removing the social norm thus boils down to reducing the time cost of children, λ , by 4.3 percent in our model.

Table 7 summarizes the results on fertility and education investment decisions. It increases fertility and reduces childlessness. It also reduces education spending. Interestingly, the effects on fertility are much smaller quantitatively in our model with the status externality (a 2.9 percent increase) than what Myong, Park, and Yi (2021) find (an 11.2 percent increase). Similarly, our childlessness rate falls only by 0.2 percentage points, while in their model it decreases from 2.9 percent to 1.3 percent. Further, in our model shutting down the social norm increases fertility the most for the lower-income quintiles, while in their model the effect is largest for high levels of mother's education. In the data, comparing different cohorts, fertility has fallen the least in the highest-income quintile (as shown in online Appendix Table A7). This is counter to the implication of their model assuming, as argued by the authors, that the social norm became more relevant over time. Further, when comparing the importance of the social norm with the status externality in our model, we find that turning off the status externality increases fertility by much more than removing the social norm.

There are two further reasons why we believe the high cost of education is more important for understanding recent fertility behaviors in Korea than social norms. The Korea Population, Health and Welfare Association conducted a survey in 2017 to understand the reasons behind the low birth rates.²⁴ The most important reason by far was the financial burden of raising a child (43 percent of respondents mention this as the main reason). The unequal division of housework and childcare toward women was also mentioned in the same survey but not ranked as a main reason by many (only 4.3 percent of respondents). See online Appendix A.8 for details.

Secondly, perceptions about the division of housework in the family have changed dramatically over time. Figure 4 shows that between 2008 and 2022, the fraction of 30- to 39-year-olds (the ages during which most births happen in Korea) who think that the wife is mostly responsible for housework has declined from 70 percent to roughly 20 percent. By now, the majority of Koreans of childbearing age believe that housework should be shared equally.²⁵ Given this dramatic change, social norms

²⁴ See *Low Fertility Awareness Survey* (Korea Population, Health and Welfare Association 2017).

²⁵ The general trend of declining perceptions of the gendered division of housework is robust across all age groups.

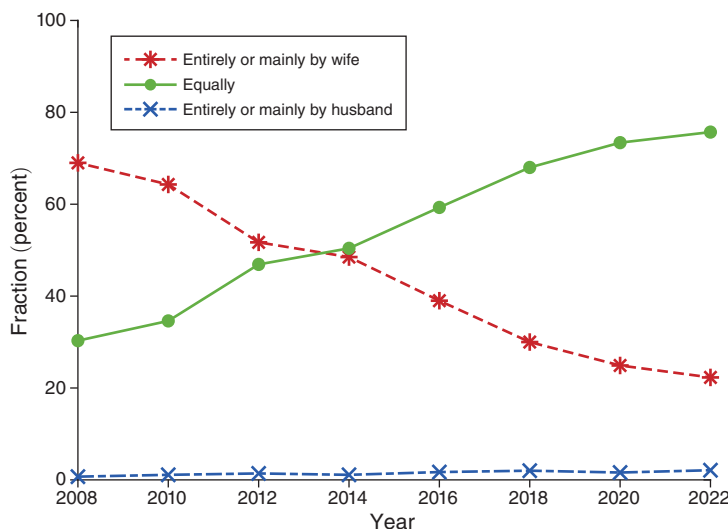


FIGURE 4. TRENDS ON PERCEPTION ABOUT HOUSEWORK DIVISION WITHIN HOUSEHOLDS

Note: The figure plots the fraction of people aged between 30 and 39 who think that housework should be (i) done entirely or mainly by wife, (ii) shared equally, or (iii) done entirely or mainly by husband.

Source: Social Survey, Statistics Korea (2022)

about the gendered division of housework are unlikely to be the main factor behind the lowest low fertility in recent years. Such norms have become much more egalitarian, which should imply, all else equal, that the distortion from the social norm should have declined, not increased, over time.

Of course, this does not mean that social norms are irrelevant. Note that the baseline sample for the model estimation in Myong, Park, and Yi (2021) consists of women 45 to 70 years old in 2015, which means those who were born in 1945–1970, while our analysis focuses on the generation born in 1970–1975, i.e., on a more recent generation. It may well be that the earlier decline in fertility was largely driven by social norms about the gendered division of housework, while the more recent decline is more related to status externalities in education.

V. Pro-natal Transfers and Education Taxes

Like many other countries, the Korean government has implemented various policies to fight falling birth rates. In particular, it initiated *The First Basic Plan on Low Fertility and Aging Society* in 2006 (Hong et al. 2016). In this section, we use our model to investigate the effect of two policies aimed at stimulating the birth rate. First, fertility could be directly stimulated by giving parents cash or in-kind transfers tied to a birth (i.e., a pro-natal transfer). Second, since the status externality leads parents to overinvest in their children's education, a tax on private education could reduce equilibrium education spending, making children cheaper and thus stimulating fertility. In what follows, we investigate the effects of these policies before turning to the more complicated question of an optimal policy in Section VI. We

focus on steady-state comparisons, which are useful for gauging the long-run implications. Yet we also analyze transitional dynamics in online Appendix B.5.

A. Pro-natal Transfers

Pro-natal transfers are an intuitive way to promote fertility, where a government benefit is tied to the number of children a family has. In Korea, some districts started introducing cash grants for births as early as 2003. By the end of 2011, most districts (229 out of 260) were providing such grants, and these programs are expected to continue to grow in size. Beyond newborns, many countries provide monetary benefits for all children below a certain age. In Germany, for example, a monthly cash transfer of about €200 per child (called *Kindergeld*) is given to parents until offspring reach the age of 18. This is essentially a pro-natal transfer, as it reduces the cost to parents of having children.

To capture such programs, we add two terms to the household budget constraints:

$$(10) \quad c + xn + T \leq wh(1 - \lambda n - l) + T_n(n),$$

where $T_n(n) > 0$ captures pro-natal transfers. In each period, the government budget is balanced by adjusting lump-sum taxes T to finance government spending:

$$(11) \quad \sum_j^{N_\kappa} \tilde{\pi}_j^\kappa \sum_i^{N_b} \pi_i^b \int [T_n(n) - T] F(dh, b_i, \kappa_j^p) = 0,$$

where $\tilde{\pi}_j^\kappa$ captures the probability mass of κ_j^p . We consider nondistortionary lump-sum taxes, as we want to focus on the issue of the status externality, while avoiding another distortion caused by taxes on income or consumption.

We consider a simple function for pro-natal transfers: $T_n(n) = \psi n$. To investigate potential nonlinearity in the policy effects, we consider two different levels of ψ : 0.01 and 0.02. A transfer of $\psi = 0.01$ corresponds to a monthly child allowance of US\$35, or 1.5 percent of monthly income per child for 18 years.²⁶

Table 8 shows that pro-natal transfers clearly increase the fertility rate. For example, with a ψ of 0.01, the new steady-state fertility rate increases by 4.2 percent. The positive impact on fertility is also observed at the extensive margin: for example, when ψ increases to 0.01 and 0.02, the childlessness rate falls to 3.1 percent and 2.9 percent, respectively. However, the required funding also increases rapidly, with 4.5 percent of output required to sustain a transfer of $\psi = 0.02$. The effects on fertility, meanwhile, are relatively small: the fertility rate only increases from 1.92 to 2.08 (or 8.5 percent).²⁷

²⁶This value is based on the assumption that annual GDP per capita is US\$28,732 (2015 estimate). Specifically, let M be the monthly payment to each child. The total transfer payments per child until the age of 18 in the data relative to GDP per capita over 25 years is $(M \times 12 \times 18)/(28,732 \times 25)$. This corresponds to $\psi/0.958$ in the model. By equating these two, we obtain the relationship between M and ψ . Finally, M divided by US\$28,732/12 gives the fraction relative to monthly income.

²⁷Interestingly, the effects of the policy are somewhat larger compared to a world without status externalities, as discussed in online Appendix B.3.

TABLE 8—LONG-RUN EFFECTS OF PRO-NATAL TRANSFERS AND EDUCATION TAXES

	Baseline	Pro-natal transfers		Education taxes	
		$\psi = 0.01$	$\psi = 0.02$	$\tau_x = 0.1$	$\tau_x = 0.2$
Fertility rate n	1.92	2.00	2.08	1.97	2.01
(Percent change)		(4.2)	(8.5)	(2.9)	(4.8)
Childlessness rate (percent)	3.4	3.1	2.9	3.3	3.3
Avg x per child/income (percent)	11.0	10.4	9.9	8.9	7.4
Income elasticity of n	0.080	0.020	−0.021	0.027	−0.002
Income elasticity of x	0.775	0.824	0.868	0.784	0.788
Avg labor supply	0.290	0.288	0.286	0.282	0.275
Avg human capital	3.29	3.24	3.20	3.18	3.11
Output per capita	0.958	0.938	0.918	0.901	0.860
Gini income	0.266	0.268	0.271	0.271	0.275
IGE	0.327	0.326	0.324	0.309	0.293
T/Y (percent)		2.1	4.5	1.9	3.2

Notes: Numbers in parentheses show the percent change relative to the baseline model. T denotes lump-sum taxes in the second and third columns (pro-natal transfer experiments), whereas it denotes lump-sum transfers in the last two columns (education tax experiments).

The fertility effects of pro-natal transfers in our model are in line with recent empirical evidence from Korea. Hong et al. (2016) estimate the causal effect of these transfers in Korea using regional and time variation, finding that a one-time cash bonus of US\$1,000 increases the crude birth rate by 4.4 percent. More recently, Kim (2020) exploits the same policy changes but uses birth outcomes over a longer time horizon, concluding that a 10 percent increase in cash transfers raised birth rates by approximately 0.4–0.6 percent, depending on birth order.²⁸ These estimates are similar in magnitude to our model. When we compare a transfer of $\psi = 0.01$ to $\psi = 0.02$, i.e., a 100 percent increase in pro-natal transfers, the completed fertility rate goes up by 4.1 percent.

As fertility increases in response to pro-natal transfers, the average spending on private education per child decreases. For instance, with $\psi = 0.02$, parents have more children but invest less per child (education spending per child declines from 11.0 percent to 9.9 percent of income). It is also worth noting that the income elasticity of n decreases and that of x increases with larger pro-natal transfers. The rise in fertility among low-income families is greater than among high-income families in response to pro-natal transfers, leading to a larger decrease in per child education spending in low-income families. Because parents invest less per child, the average human capital and output per capita are lower in the new steady state.²⁹

A pro-natal transfer also leads to a moderate long-run increase in inequality, reflected in a higher Gini coefficient. This is due to the reduced education spending by poorer parents relative to richer parents when a pro-natal transfer is introduced.

²⁸The relatively large estimates of Hong et al. (2016) may partly be caused by a change in the timing of births, making the lower estimates of Kim (2020) more relevant here.

²⁹We have also considered means-tested pro-natal transfers, which is somewhat more effective in raising fertility, especially among poor parents. Accordingly, average labor supply, human capital, and output per capita fall by more. See online Appendix B.4 for details.

B. Private Education Investment Tax

As shown in Section IV, low fertility is partially caused by the status externality, which leads to high education spending. One might therefore consider taxing education. While most countries subsidize rather than tax education, the Korean government has long struggled to reduce the high demand for private education. In 1980, the government entirely prohibited private education through *hagwons* and private tutoring, although this was later declared unconstitutional. Nonetheless, there remain restrictions on the hours of operation of *hagwons*, as detailed in online Appendix A.3. Similarly, the Chinese government has recently introduced severe regulations in private education industries to fight falling birth rates.³⁰

We now investigate the effects of an education tax in our model. We extend the budget constraint of the household as follows:

$$(12) \quad c + (1 + \tau_x)xn \leq wh(1 - \lambda n - l) + T,$$

where $\tau_x \in [0, 1]$ denotes tax on private education investment. We consider two different levels of τ_x : 0.1 and 0.2. Again, we require the government budget to balance in each period, which is achieved through lump-sum transfers T to consumers:

$$(13) \quad \sum_j^{N_\kappa} \tilde{\pi}_j^\kappa \sum_i^{N_b} \pi_i^b \int (\tau_x xn - T) F(dh, b_i, \kappa_j^p) = 0.$$

The last two columns of Table 8 summarize the results. We find that the education tax is quite powerful in reducing parental investment. Private education expenditures relative to income drop considerably, from 11.0 percent to 8.9 percent and 7.4 percent for education tax rates of 10 percent and 20 percent, respectively. A parent's lower demand for quality implies a higher demand for the quantity of children: the fertility rate increases when an education tax is introduced. As parents choose to invest less per child, average human capital in the long run becomes lower, and parents work less. Together, these lead to lower output per capita in the long run.

In sum, both pro-natal transfers and taxes on private education are found to be effective at raising fertility, although their effects are quantitatively moderate. Yet we also find that these policies necessarily involve intergenerational conflicts since future generations suffer from lower human capital and thus lower output per capita.

VI. Optimal Policy

In the previous section, we saw that pro-natal transfers and education taxes affect fertility and education decisions. We have also seen that policies that lower education naturally reduce the human capital of future generations and thus reduce output per capita in the future. But how do these policies impact welfare? What is the optimal policy?

³⁰ See "China Bans For-Profit Tutoring In Reforms Aimed At Boosting The Birth Rate" (Graison Dangor, *Forbes*, July 24, 2021).

While these are obvious questions to ask, answering them is far from straightforward. In this section, we begin by discussing why this problem is nontrivial in our heterogeneous agent model with endogenous fertility and then explain how we address this challenge. Then, we investigate the optimal mix of the two policy instruments considered in the previous section.³¹

A. Welfare Concepts and the Planning Problem

Externalities generally lead to distortions. In our model, parents do not consider the effect of their education investment on other parents' children. Thus, the marginal private return from education is higher than the marginal social return, and parents will overinvest in education.³² Loosely speaking, the equilibrium allocation will thus not be efficient. One would then like to know what the optimal allocation is that a planner would choose. While this logic seems clear and simple, it entails a number of complications. The notion of Pareto Efficiency is not well defined in models with endogenous fertility that imply changing sets of people across allocations. Naturally, then, there is no planning problem that recovers the first-best allocation or, as is often the case, all allocations on the contract curve.

A concept closely related to Pareto Efficiency defined for models with endogenous fertility is *A-Efficiency*, first proposed by Golosov, Jones, and Tertilt (2007). An allocation is considered *A-Efficient* if there is no other allocation that is weakly preferred by all people alive in both allocations and strictly preferred by at least one person alive in both allocations. Thus, *A-Efficiency* is a natural modification of Pareto Efficiency, which focuses only on those alive in both allocations whenever comparing two allocations.³³ Applying this concept to our context, note that the logic is similar to the pollution example given by Golosov, Jones, and Tertilt (2007). Their example shows that when consumption affects other people negatively, the equilibrium allocation features higher consumption and population levels than optimal. Furthermore, taxes on consumption and children can address the externality and implement the first-best allocation. But how does one find such *A-Efficient* allocations? Golosov, Jones, and Tertilt (2007) prove that if the solution to a planning problem that maximizes the weighted sum of utility of the first generation is unique (and all weights are strictly positive), then the allocation is *A-Efficient*. In other words, we can recover many different *A-Efficient* allocations that maximize the weighted sum of the first generation with varying weights.

An additional consideration is that a large part of welfare gains across policies in heterogeneous agent models typically originate from redistributing resources from the rich to the poor. While welfare gains through redistribution are of interest in a general sense, they are not the focus of our paper. Rather, we are interested in

³¹ One might consider richer policy tools, such as progressive taxation. This is left for future work.

³² Of course, there might also be other externalities not present in our framework that lead social returns to be higher than private ones, such as the idea spillovers considered by Jones (2022).

³³ This concept ignores the "views" of additional people whenever one allocation has a higher fertility rate and ignores the "views" of those who do not come into existence when considering an alternative allocation with a smaller population. Since preferences of nonexistent people are hard to define and impossible to measure, simply ignoring them is a pragmatic way to move forward.

the effects of a particular distortion.³⁴ To isolate the welfare implications of the distortion from redistributive concerns, we use Negishi weights. Negishi weights put a greater weight on rich people (in our case, those with high human capital) so that in an economy without distortions the planning problem simply recovers the equilibrium allocation.³⁵ Finally, rather than computing the unconstrained optimum, we follow the Ramsey tradition and allow the planner to use only a certain set of instruments, namely education taxes and pro-natal transfers.³⁶

With these preliminary considerations, let us now define a planning problem for our model. The planner maximizes the Negishi-weighted sum of the utility of the initial generation subject to the allocation being implementable as an equilibrium with education taxes and pro-natal transfers.³⁷ In each period following the initial reform period, we constrain the planner to impose the same taxes and transfers on everyone and to balance the government budget constraint. The planner considers utilities along the transition path rather than comparing steady states.³⁸ The baseline exercise considers permanent policy changes as in the previous section, but we also consider temporary policy changes, as shown in online Appendix B.9. For a formal definition of the planning problem, see online Appendix B.6.

B. Optimal Policy Results

We find that the optimal policy—which maximizes the Ramsey planning problem as described above—is a 21.6 percent education tax and moderate cash transfers with a ψ of 0.0175.³⁹ This amounts to a monthly cash transfer of US\$61 (2.5 percent of average income) for 18 years, according to the conversion of ψ into the monthly child allowance in Section VA. Relative to the baseline economy, the optimal policy increases the Negishi-weighted average utility of the first generation by 0.0126. To interpret the size of the welfare gain, we compute the change in consumption of the average agent with two children in the baseline equilibrium that is needed to achieve the same utility gain. This corresponds to a 1.2 percent increase in lifetime

³⁴ Welfare gains through redistribution can be easily achieved in heterogeneous agent models. Yet these “welfare gains” are usually not Pareto-improving, since the rich are made worse off, unless one takes a veil-of-ignorance perspective. See Davila et al. (2012) for further discussion.

³⁵ Negishi weights are frequently used in environmental economics (Nordhaus and Yang 1996). In contrast, the quantitative macroeconomics literature typically uses equal weights, mixing welfare gains from redistribution and those arising from correcting distortions. Notable exceptions are Domeij and Klein (2013); Heathcote, Storesletten, and Violante (2017); and Guner, Kaygusuz, and Ventura (2020). We present optimal policy results with equal welfare weights in online Appendix B.9.

³⁶ The first-best planning allocations that maximize the welfare of the first generation are difficult to compute in our model due to the discrete choice of fertility. Specifically, since no tractable optimality condition can be derived and human capital is a continuous variable, it requires the planner to compare an infinite number of combinations of fertility choices across heterogeneous agents.

³⁷ Online Appendix B.7 provides details about how we construct the Negishi weights. To check whether our Negishi weights are constructed correctly, we also solved the same planning problem for a modified economy without externality feedback, as described in online Appendix B.3. With the modified economy, we find that the optimal τ_x and ψ are zero. This shows that our attempt to separate the distortion from redistributive concerns was successful.

³⁸ Simply comparing steady states with different policies would give misleading results since this would compare allocations with different initial levels of human capital, ignoring the transition to reach the new level of human capital.

³⁹ To verify that the planner’s problem is well behaved, we plot the utility of the planner for each instrument separately in online Appendix B.8. The value varies smoothly in each policy instrument.

TABLE 9—HETEROGENEOUS EFFECTS OF THE OPTIMAL POLICY ON THE FIRST GENERATION

Average		Income quintile					All
		1st	2nd	3rd	4th	5th	
Fertility, n	Baseline	1.78	1.89	1.94	1.99	1.97	1.92
	Optimal	2.19	2.19	2.12	2.09	2.03	2.12
	Percent change	+23.2	+15.6	+9.1	+5.4	+2.8	+10.8
Childlessness rate (percent)	Baseline	5.2	2.9	3.1	3.0	2.7	3.4
	Optimal	2.3	2.2	2.9	2.9	3.0	2.7
	p.p. change	-2.9	-0.7	-0.2	-0.1	+0.3	-0.7
Investment per child, x (if $n > 0$)	Baseline	0.061	0.080	0.096	0.116	0.175	0.106
	Optimal	0.033	0.046	0.057	0.072	0.113	0.064
	Percent change	-45.5	-42.9	-40.3	-37.7	-35.6	-39.3

consumption, suggesting that noticeable welfare gains for the first generation are possible through the introduction of the optimal policy.

Table 9 shows the effect of the policy on the first generation and how the resulting allocation differs from the baseline equilibrium. Under the optimal policy, average fertility is higher by 10.8 percent, the childlessness rate is reduced by more than half a percentage point, and educational investments decrease by 39 percent. The effects are heterogeneous along the income distribution. Fertility increases the most (23 percent) for the poorest quintile and the least (3 percent) for the richest quintile. This is intuitive since it is the poor who are most affected by the externality. Similarly, the childlessness rate decreases the most for the poorest quintile. Education spending also declines the most for the poor.

How does the optimal policy affect future generations? Figure 5 shows the transitional dynamics of key variables to the new steady state with the optimal policy that is introduced unexpectedly and permanently in period 1. As discussed above, the optimal policy increases the fertility rate and decreases education spending. As a result, human capital and output decrease gradually over time. However, parents (the first generation) benefit from the optimal policy by enjoying more leisure and consumption. The last panel shows that the welfare of the first generation increases, while all future generations experience lower average utilities.⁴⁰ The reason for the welfare loss of future generations is twofold. First, in our model, only the parents—i.e., not the children—face the human capital externality. Secondly, our model considers education to be truly productive. If either of these two assumptions was changed, it should be possible to construct allocations that truly benefit every generation. To understand why, consider a status externality that leads parents to invest in improving signals for their children without truly educating them more. For example, a large part of preparation for a national entrance exam might only improve test-taking skills without increasing children's knowledge. If so, then less education spending will not lower the human capital of children and, accordingly,

⁴⁰The same is still true in the case of a temporary policy; see online Appendix Figure A10.

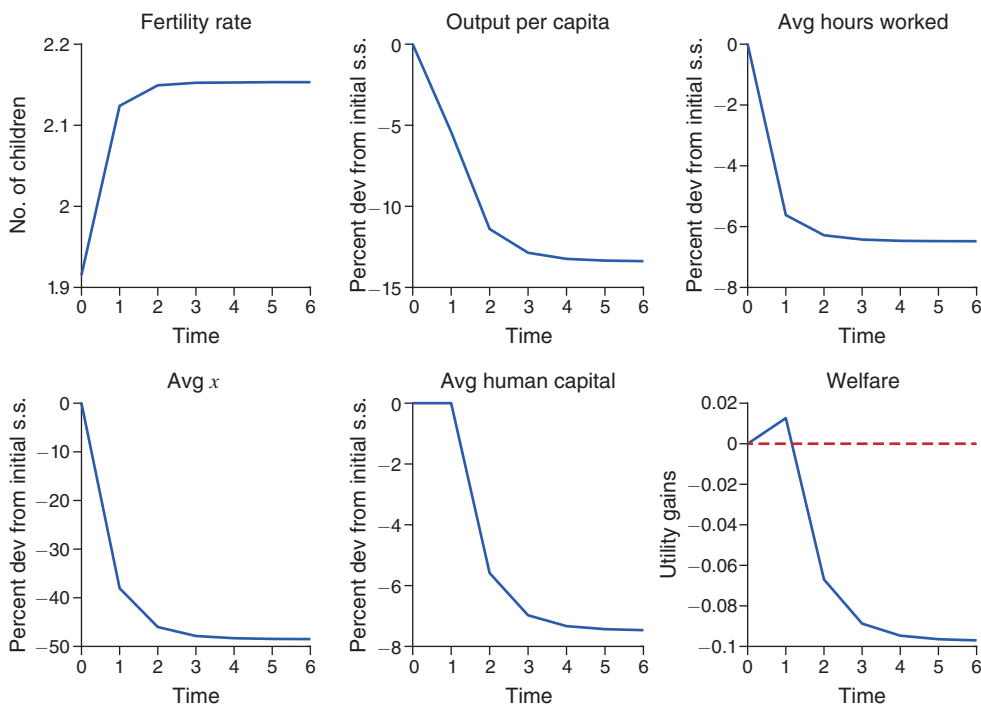


FIGURE 5. TRANSITION EQUILIBRIUM UNDER THE OPTIMAL POLICY

Notes: The optimal policy reform is introduced unexpectedly and permanently in period 1. Welfare is measured by Negishi-weighted average utility. A model period corresponds to 25 years.

not harm their welfare.⁴¹ Alternatively, suppose children faced the same externality as the parents; i.e., they cared about own consumption relative to their peers, similar to the original pollution example in Golosov, Jones, and Tertilt (2007). In such a world, it should be possible to construct a dominating allocation with lower education and consumption for future generations that nevertheless improves their welfare. Finally, note that the population size of future generations becomes larger under the policy that maximizes the first generation's welfare. Alternative views of the optimal population problem (Dasgupta 1969; de la Croix and Doepke 2021) could put weights on population size per se (something that our concept of *A-Efficiency* is silent on), which may offset the lower average utility of future generations.

VII. Tournament Model

We have modeled status concerns as a pure utility externality. A question is whether there is a deeper cause behind such an externality. One possibility is that

⁴¹The signaling value of education, which is costly but does not necessarily improve productivity, was first studied by Spence (1973). Several authors have formalized the point that investment into costly signals out of status concerns can lead to inefficient investments and even poverty; see, for example, Ireland (1998) and Moav and Neeman (2012).

status concerns capture capacity constraints in the national education system in a reduced-form way. If children compete for a limited number of high-quality universities, parents will have a strong incentive to invest in private education to improve their children's chance to get in (e.g., a rug rat race, as noted by Ramey and Ramey 2010).

We now investigate this idea by changing the model as follows. Let $\Pi(h'; \tilde{h}')$ denote the probability of getting into college. It is a function of own human capital but also others' human capital since if others invest more, the cutoff for getting in must be higher, with limited slots. We change the parental utility function so that parents derive utility specifically from their children getting into college,

$$(14) \quad U(c, l, n, h', \tilde{h}') = b \log\left(\frac{c}{\Lambda(n)}\right) + \nu \frac{l^{1-\gamma}}{1-\gamma} + \phi(n) \Pi(h'; \tilde{h}').$$

There are two key features required for Π . First, the probability of going to college increases with greater human capital: $\partial \Pi / \partial h' > 0$. Second, a greater h' makes it more difficult for others to get into college. To capture this externality, an aggregate statistic of human capital $\tilde{h}'$ enters negatively: $\partial \Pi / \partial \tilde{h}' < 0$. Note that the second property works similarly to the status externality in our baseline model. However, a key difference is that we now explicitly incorporate the possibility that some families might face very low returns to private education investments if their children have human capital that is sufficiently below some threshold level required for college entrance.⁴² To do so, we consider the following functional form for Π :

$$(15) \quad \Pi(h'; \tilde{h}') = \max\left\{0, 1 - e^{-\iota_2(h' + \iota_0 - \iota_1 \tilde{h}')}\right\}.$$

This function permits the possibility of having zero chance to go to college for low human capital, which is shaped by ι_0 . The speed at which the probability increases with human capital is controlled by ι_2 . It thus determines the equilibrium share of college graduates. The importance of relative performance for getting into college is governed by ι_1 , similar to χ in our status externality model. Finally, the function approaches one asymptotically so that children with high enough human capital are essentially guaranteed a college slot. To calibrate this modified model, we add the share of college graduates as a new data target. The share of tertiary-educated 25- to 34-year-olds was 69 percent in 2021.⁴³ We keep the other target moments as in the status externality model.⁴⁴ Generally, the tournament model also fits the data well, as shown in online Appendix Table A13.

As expected, the tournament model generates a portion of people who face a zero probability of their child going to college.⁴⁵ This is concentrated among the

⁴² While we do not model the threshold explicitly, we later verify in the quantitative model that the total number of children in college indeed stays roughly constant in response to policy changes. Thus, our formulation captures well the idea of college slot scarcity.

⁴³ See *Education at a Glance 2022: OECD Indicators*.

⁴⁴ $\tilde{h}'$ is set to be the top 15 percent average because we still target the investment spillover estimate based on the top 15 percent as in the status externality model. Since we added two new parameters but only one new moment, the model is now exactly identified.

⁴⁵ Recall that parents do not know their children's ability with certainty before they make the fertility choice. Thus, it is possible to have kids with zero chance of going to college. A full histogram of college probabilities is shown in online Appendix Figure A14.

TABLE 10—LONG-RUN EFFECTS OF POLICIES: TOURNAMENT MODEL

	Baseline	Pro-natal transfers		Education taxes	
		$\psi = 0.01$	$\psi = 0.02$	$\tau_x = 0.1$	$\tau_x = 0.2$
Fertility rate n	1.90	1.97	2.08	1.89	1.88
(Percent change)		(4.1)	(9.5)	(−0.5)	(−1.0)
Childlessness rate (percent)	2.6	1.9	1.3	2.7	2.8
Avg x per child/income (percent)	9.6	9.4	9.0	8.7	7.9
Income elasticity of n	0.084	0.020	−0.054	0.076	0.067
Income elasticity of x	0.673	0.736	0.803	0.694	0.714
Avg labor supply	0.308	0.307	0.305	0.304	0.300
Avg human capital	3.07	3.01	2.94	2.96	2.87
Output per capita	0.939	0.919	0.892	0.897	0.859
Gini income	0.223	0.229	0.237	0.232	0.240
IGE	0.293	0.309	0.323	0.297	0.300
T/Y (percent)		2.1	4.7	1.7	3.0

Notes: Numbers in parentheses show the percent change relative to the baseline model. T denotes lump-sum taxes in the second and third columns (pro-natal transfer experiments), whereas it denotes lump-sum transfers in the last two columns (education tax experiments).

poorer parents. One could thus expect policies to affect parents along the income distribution quite differently from our status externality model. In particular, if poor parents' children have no intention to attend college in any case, then the general equilibrium effect that a policy has on $\tilde{h}$ may not be operative for them. This could lead to policy affecting parents along the income distribution in opposite ways from the main model. We now explore this possibility quantitatively.

Table 10 is the analog of Table 8 from our status externality model. While the effect of pro-natal transfers is very similar, the effect of education taxes is quite different. While in the status externality model fertility increased with the policy, it is hardly affected in the tournament model. The reason is that an education tax always has two effects: it lowers the externality, but it also makes education more expensive. In the tournament model the two effects roughly cancel out so that fertility does not change much. Further, we find that the expected effect that poorer parents are less responsive to an education tax does not materialize. Rather, the policy does cause parents in the poorest quintile to reduce education spending by the most (31 percent) and in the richest quintile the least (23 percent). The reason is that given the high college rate in Korea, even among the poorest quintile of parents a large share of children do attend college. So while some parents are below a threshold level that makes them inelastic to policy, this is only a small fraction of parents even within the poorest quintile and hence quantitatively not important.

Next, we compare optimal policies in the two model versions. Sticking to the same set of instruments as before, we find that the optimal intervention is larger in the tournament model, which requires an education tax $\tau_x = 36.3$ percent and a pro-natal transfer $\psi = 0.0244$ (compared to 21.6 percent and 0.0175 in the status externality model, respectively). The welfare benefit of the optimal policy on the first generation is also slightly higher than in the status externality model. The heterogeneous effects on the first generation are quite similar, as shown in Table 11. In particular, as before, the investment per child does decrease the most for the poorest quintile, and the slope is quite similar compared to Table 9. Similarly, fertility increases the most for the poorest quintile and the least for the richest quintile.

TABLE 11—HETEROGENEOUS EFFECTS OF THE OPTIMAL POLICY ON THE FIRST GENERATION:
TOURNAMENT MODEL

Average		Income quintile					All
		1st	2nd	3rd	4th	5th	
Fertility, n	Baseline	1.71	1.92	1.99	1.98	1.87	1.90
	Optimal	2.30	2.18	2.14	2.05	1.93	2.12
	Percent change	+34.0	+13.3	+7.9	+4.0	+2.8	+11.9
Childlessness rate (percent)	Baseline	5.5	1.8	1.4	1.6	2.5	2.6
	Optimal	0.5	0.9	1.1	1.1	2.4	1.2
	p.p. change	-5.0	-1.0	-0.3	-0.5	-0.1	-1.3
Investment per child, x (if $n > 0$)	Baseline	0.054	0.081	0.093	0.105	0.120	0.091
	Optimal	0.029	0.048	0.059	0.067	0.078	0.056
	Percent change	-45.8	-40.3	-37.1	-35.9	-34.7	-38.1
No. of children going to college	Baseline	1.15	1.45	1.54	1.56	1.53	1.45
	Optimal	1.36	1.53	1.57	1.53	1.50	1.50
	Percent change	+18.5	+5.9	+1.5	-1.6	-1.8	+3.7

Table 11 also shows how the pattern of college enrollment changes. Since fertility changes with policy, which also affects the chance to get into college, we report the number of children per family who go to college rather than the probability. The table confirms that, as desired, the total number of children per family in college remains roughly the same. However, families along the income distribution react differently. With the optimal policy in place, poorer families on average send more children to college, while richer families send fewer children to college.⁴⁶

In sum, we find that pro-natal transfers affect fertility and education similarly in the tournament model as in the benchmark model. In contrast, education taxes, which are powerful in the externality model, have little impact, and the welfare benefits of the optimal policy are somewhat larger in the tournament model. Moreover, if the main reason for the spillover effects was indeed college slot scarcity, then one might also consider alternative policy levers, such as increasing the number of college slots. A serious analysis of such alternative policies would require including higher education and its funding into the model. This is beyond the scope of this paper and left for future research. Clearly, this is an important avenue for future work since whether the externality originates in the higher education system or is based on social norms is an open question we cannot answer in this paper.

VIII. Concluding Remarks

In this paper, we present a heterogeneous agent model of endogenous fertility with a concern for the relative quality of children. Our model enriches the standard quality-quantity model and can account for various cross-sectional patterns of

⁴⁶Online Appendix Figure A14 (panel B) shows the college probability as a function of child human capital. Compared to the baseline equilibrium, the threshold below which children have a zero probability moves to the left under the optimal policy, and all children with human capital above the threshold now have a slightly higher probability. Yet in equilibrium, average human capital falls, leading to lower college enrollment. This is undone with the higher fertility rate, with some of the additional children filling up the freed-up college slots.

fertility and education investment in Korea. In our model, the absence of a status externality increases fertility by 28 percent, driven in particular by low-income families.

We investigate the transition of economic variables and the welfare of different generations after various policy reforms. In the literature, the effect of pro-natal policies on fertility is typically analyzed empirically. Our approach uses a dynamic equilibrium model framework, which allows us to study the dynamic effect on other variables over time and the welfare of different generations. This is important because the distortion from the status externality complicates the problem by affecting various decisions such as education investment and labor supply in addition to fertility decisions. This, in turn, leads to differential benefits for different generations following policy reforms. Indeed, we find that the optimal policy from the perspective of the current generation, which consist of pro-natal transfers and education taxes, may lead to undesirable outcomes for future generations and could cause a conflict between current and future generations. In addition, our policy experiments reveal the heterogeneous effects of policy reforms along the income distribution. For instance, the optimal policy disproportionately increases fertility and decreases education investment among low-income households.

It is worth pointing out some caveats and directions for future research. First, as is always the case, results depend on modeling assumptions. We focus on human capital transmission and abstract from wealth transfers since bequests are rare in Korea.⁴⁷ Bequests could be an alternative way for parents to provide status to their children. Taxes on bequests are high in Korea, which might be another reason parents invest in human capital during childhood. If so, removing the bequest tax distortion could potentially lower private education investments. Another thing left out is parental time in the human capital production function. There is a large literature documenting the importance of parental time. Adding it might amplify or dampen the role of the externality depending on the degree of substitutability between time and monetary investments (Yum 2023). We also abstract from public education as a policy lever. Allowing the government to directly choose public education inputs financed through taxes might be another interesting avenue to pursue. Such analyses should ideally be investigated in a more general framework where substitutability between public and private education is flexible, given its importance for aggregate and distributional effects of public education (Jang and Yum forthcoming).

Second, assessing optimal policy necessarily depends on the optimality concept, which is not straightforward in models with endogenous fertility choice. We have focused on maximizing the utility of the first generation, but one could of course consider other concepts that put more weight on future generations. Third, our current paper cannot empirically identify whether college slot scarcity is the driver behind the status externality. If so, other policy levers might be available, such as increasing the number of college slots or changing admission policies. Given that almost 70 percent of Koreans attend college, the real issue might rather be the scarcity of slots in elite universities. To assess this possibility, a model with colleges

⁴⁷ Kim (2010) finds that the fraction of households that ever received a bequest is only around 3 percent, much smaller than in the United States (28 percent).

of different qualities would be needed (e.g., Kang 2023). Adding a more explicit higher education sector would be an interesting extension. Finally, while we argued that externalities are the most important driver for the low fertility rate of women born in the 1970s, social norms about the gendered division of housework may have played an important role for other cohorts. Carefully disentangling the role of the two mechanisms for different time periods would be an interesting avenue to pursue. We believe that these offer promising avenues for future work.

REFERENCES

- Aaronson, Dan, Fabian Lange, and Bhash Mazumder. 2014. "Fertility Transitions Along the Extensive and Intensive Margin." *American Economic Review* 104 (11): 3701–24.
- Abel, Andrew B. 1990. "Asset Prices under Habit Formation and Catching up with the Joneses." *American Economic Review* 80 (2): 38–42.
- Akerlof, George. 1976. "The Economics of Caste and of the Rat Race and Other Woeful Tales." *Quarterly Journal of Economics* 90 (4): 599–617.
- Anderson, Thomas M., and Hans-Peter Kohler. 2013. "Education Fever and the East Asian Fertility Puzzle: A Case Study of Low Fertility in South Korea." *Asian Population Studies* 9 (2): 196–215.
- Ashraf, Nava, Erica Field, and Jean Lee. 2014. "Household Bargaining and Excess Fertility: An Experimental Study in Zambia." *American Economic Review* 104 (7): 2210–37.
- Bar, Michael, Moshe Hazan, Oksana Leukhina, David Weiss, and Hosny Zoabi. 2018. "Why did Rich Families Increase their Fertility? Inequality and Marketization of Child Care." *Journal of Economic Growth* 23 (4): 427–63.
- Barro, Robert J., and Gary S. Becker. 1989. "Fertility Choice in a Model of Economic Growth." *Econometrica* 57 (2): 481–501.
- Baudin, Thomas, David de la Croix, and Paula E. Gobbi. 2015. "Fertility and Childlessness in the United States." *American Economic Review* 105 (6): 1852–82.
- Becker, Gary S. 1960. "An Economic Analysis of Fertility." In *Demographic and Economic Change in Developed Countries*, edited by George B. Roberts, 209–40. New York, NY: Columbia University Press.
- Becker, Gary S., and Robert J. Barro. 1988. "A Reformulation of the Theory of Fertility." *Quarterly Journal of Economics* 103 (1): 1–25.
- Becker, Gary S., and Nigel Tomes. 1976. "Child Endowments and the Quantity and Quality of Children." *Journal of Political Economy* 84 (4): S143–62.
- Bertrand, Marianne, and Adair Morse. 2016. "Trickle-down Consumption." *Review of Economics and Statistics* 98 (5): 863–79.
- Billari, Francesco C., and Hans-Peter Kohler. 2004. "Patterns of Low and Lowest-Low Fertility in Europe." *Population Studies* 58 (2): 161–76.
- Cameron, A. Colin, Jonah B. Gelbach, and Douglas L. Miller. 2008. "Bootstrap-Based Improvements for Inference with Clustered Errors." *Review of Economics and Statistics* 90 (3): 414–27.
- Cavalcanti, Tiago, Georgi Kocharkov, and Cezar Santos. 2020. "Family Planning and Development: Aggregate Effects of Contraceptive Use." *Economic Journal* 131 (634): 624–57.
- Chetty, Raj, Nathaniel Hendren, Patrick Kline, and Emmanuel Saez. 2014. "Where is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States." *Quarterly Journal of Economics* 129 (4): 1553–1623.
- Choi, Eleanor Jawon, and Jisoo Hwang. 2020. "Transition of Son Preferences: Evidence from South Korea." *Demography* 57 (2): 627–52.
- Choi, Hoon, and Álvaro Choi. 2016. "Regulating Private Tutoring Consumption in Korea: Lessons from Another Failure." *International Journal of Educational Development* 49: 144–56.
- Cordoba, Juan Carlos, Xiyang Liu, and Marla Ripoll. 2016. "Fertility, Social Mobility and Long Run Inequality." *Journal of Monetary Economics* 77: 103–24.
- Daruich, Diego, and Julian Kozłowski. 2020. "Explaining Intergenerational Mobility: The Role of Fertility and Family Transfers." *Review of Economic Dynamics* 36: 220–45.
- Dasgupta, Partha. 1969. "On the Concept of Optimal Population." *Review of Economic Studies* 36 (3): 295–381.
- Davila, Julio, Jay H. Hong, Per Krusell, and Jose-Victor Rios-Rull. 2012. "Constrained Efficiency in the Neoclassical Growth Model with Uninsurable Idiosyncratic Shocks." *Econometrica* 80 (6): 2431–67.

- de la Croix, David, and Matthias Doepke.** 2003. "Inequality and Growth: Why Differential Fertility Matters." *American Economic Review* 93 (4): 1091–1113.
- de la Croix, David, and Matthias Doepke.** 2021. "A Soul's View of the Optimal Population Problem." *Mathematical Social Sciences* 112: 98–108.
- de Silva, Tiloka.** 2021. "The Impact of Private Tutoring on Higher Education Outcomes: Evidence from South Korea." In *Workplace Productivity and Management Practices*, edited by Solomon W. Polachek, Konstantinos Tatsiramos, Gionanni Russo, and Gijs van Houten, 239–76. Bingley, UK: Emerald Publishing Limited.
- Doepke, Matthias, Anne Hannusch, Fabian Kindermann, and Michèle Tertilt.** 2023. "The Economics of Fertility: A New Era." In *Handbook of the Economics of the Family*, Vol. 1, edited by Shelly Lundbeg and Alessandra Voena, 151–254. Amsterdam: Elsevier.
- Doepke, Matthias, and Fabian Kindermann.** 2019. "Bargaining over Babies: Theory, Evidence, and Policy Implications." *American Economic Review* 109 (9): 3264–3306.
- Domeij, David, and Paul Klein.** 2013. "Should Day Care be Subsidized?" *The Review of Economic Studies* 80 (2): 568–95.
- Easterlin, Richard A.** 1966. "On the Relation of Economic Factors to Recent and Projected Fertility Changes." *Demography* 3 (1): 131–53.
- Fagereng, Andreas, Magne Mogstad, and Marte Rønning.** 2021. "Why do Wealthy Parents Have Wealthy Children?" *Journal of Political Economy* 129 (3): 703–56.
- Ferrer-i-Carbonell, Ada.** 2005. "Income and Well-Being: An Empirical Analysis of the Comparison Income Effect." *Journal of Public Economics* 89 (5): 997–1019.
- Gali, Jordi.** 1994. "Keeping Up with the Joneses: Consumption Externalities, Portfolio Choice, and Asset Prices." *Journal of Money, Credit and Banking* 26 (1): 1–8.
- Genicot, Garance, and Debraj Ray.** 2017. "Aspirations and Inequality." *Econometrica* 85 (2): 489–519.
- Golosov, Mikhail, Larry E. Jones, and Michèle Tertilt.** 2007. "Efficiency with Endogenous Population Growth." *Econometrica* 75 (4): 1039–71.
- Greenwood, Jeremy, and Ananth Seshadri.** 2002. "The U.S. Demographic Transition." *American Economic Review* 92 (2): 153–59.
- Greenwood, Jeremy, Ananth Seshadri, and Guillaume Vandenbroucke.** 2005. "The Baby Boom and Baby Bust." *American Economic Review* 95 (1): 183–207.
- Guner, Nezih, Ezgi Kaya, and Virginia Sánchez-Marcos.** Forthcoming. "Labor Market Institutions and Fertility." *International Economic Review*.
- Guner, Nezih, Remzi Kaygusuz, and Gustavo Ventura.** 2020. "Child-Related Transfers, Household Labour Supply, and Welfare." *Review of Economic Studies* 87 (5): 2290–2321.
- Guryan, Jonathan, Erik Hurst, and Melissa Kearney.** 2008. "Parental Education and Parental Time with Children." *Journal of Economic Perspectives* 22 (3): 23–46.
- Hannusch, Anne.** 2019. "Taxing Families: The Impact of Child-related Transfers on Maternal Labor Supply." Unpublished.
- Heathcote, Jonathan, Kjetil Storesletten, and Giovanni L. Violante.** 2017. "Optimal Tax Progressivity: An Analytical Framework." *Quarterly Journal of Economics* 132 (4): 1693–1754.
- Hong, Sok Chul, Young Il Kim, Jae Young Lim, and Mee Young Yeo.** 2016. "Pro-Natalist Cash Grants and Fertility: A Panel Analysis." *Korean Economic Review* 32 (2): 331–54.
- Ireland, Norman J.** 1998. "Status-Seeking, Income Taxation and Efficiency." *Journal of Public Economics* 70 (1): 99–113.
- Jang, Youngsoo, and Minchul Yum.** Forthcoming. "Aggregate and Intergenerational Implications of School Closures: A Quantitative Assessment." *American Economic Journal: Macroeconomics*.
- Jenkins, David.** 2016. "Conspicuous consumption in the United States and China." *Journal of Economic Behavior & Organization* 127: 115–32.
- Jones, Charles I.** 2022. "The End of Economic Growth? Unintended Consequences of a Declining Population." *American Economic Review* 112 (11): 3489–3527.
- Jones, Larry E., Alice Schoonbroodt, and Michèle Tertilt.** 2010. "Fertility Theories: Can They Explain the Negative Fertility-Income Relationship?" In *Demography and the Economy*, edited by John B. Shoven, 43–100. Chicago, IL: University of Chicago Press.
- Jones, Larry E., and Michèle Tertilt.** 2008. "An Economic History of Fertility: US 1826–1960." In *Frontiers of Family Economics*. Vol. 1, edited by Peter Rupert, 165–230. Bingley, UK: Emerald Group Publishing Limited.
- Kang, Hyunjae.** 2023. "Parental Investment, Child's Efforts, and Intergenerational Mobility." PhD diss. State University of New York at Stony Brook.
- Kim, Hisam.** 2010. *Intergenerational Transfers and Old-Age Security in Korea*. Chicago, IL: University of Chicago Press.

- Kim, Seongeun, Michèle Tertilt, and Minchul Yum.** 2024. "Replication data for: Status Externalities in Education and Low Birth Rates in Korea." American Economic Association [publisher], Inter-university Consortium for Political and Social Research [distributor]. <https://doi.org/10.3886/E195662V1>.
- Kim, Sunwoong, and Ju-Ho Lee.** 2010. "Private Tutoring and Demand for Education in South Korea." *Economic Development and Cultural Change* 58 (2): 259–96.
- Kim, Wookun.** 2020. "Baby Bonus, Fertility, and Missing Women." California Center for Population Research Working Paper 2020-008.
- Korea Labor Institute.** 2018. "Korean Labor and Income Panel Study, 1998-2018." https://www.kli.re.kr/boardDownload.es?bid=0019&list_no=133256&seq=13813 (accessed August 12, 2019).
- Korea Population, Health and Welfare Association.** 2017. "Low Fertility Awareness Survey." <https://www.ppfk.or.kr/data-board/report/pp-survey-detail/3> (accessed February 27, 2023).
- Lee, R. D., and T. Miller.** 1990. "Population Growth, Externalities to Childbearing, and Fertility Policy in Developing Countries." In *Proceedings of the World Bank Annual Conference on Development Economics*. 275–304. Washington, DC: World Bank.
- Lee, Sam-Silk.** 2009. "Low Fertility and Policy Responses in Korea." *Japanese Journal of Population* 7 (1): 57–70.
- Ljungqvist, Lars, and Harald Uhlig.** 2000. "Tax Policy and Aggregate Demand Management Under Catching Up with the Joneses." *American Economic Review* 90 (3): 356–66.
- Ma, Li.** 2016. "Female Labour Force Participation and Second Birth Rates in South Korea." *Journal of Population Research* 33: 173–95.
- Manuelli, Rodolfo E., and Ananth Seshadri.** 2009. "Explaining International Fertility Differences." *Quarterly Journal of Economics* 124 (2): 771–807.
- Moav, Omer, and Zvika Neeman.** 2012. "Saving Rates and Poverty: The Role of Conspicuous Consumption and Human Capital." *Economic Journal* 122 (563): 933–56.
- Myong, Sunha, Jungjae Park, and Junjian Yi.** 2021. "Social Norms and Fertility." *Journal of the European Economic Association* 19 (5): 2429–66.
- Nakamura, Emi, and Jón Steinsson.** 2018. "Identification in Macroeconomics." *Journal of Economic Perspectives* 32 (3): 59–86.
- Nordhaus, William D., and Zili Yang.** 1996. "A Regional Dynamic General-Equilibrium Model of Alternative Climate-Change Strategies." *American Economic Review* 86 (4): 741–65.
- Podoshen, Jeffrey S., Lu Li, and Junfeng Zhang.** 2011. "Materialism and Conspicuous Consumption in China: A Cross-Cultural Examination." *International Journal of Consumer Studies* 35 (1): 17–25.
- Psacharopoulos, George, and Harry Antony Patrinos.** 2018. "Returns to Investment in Education: A Decennial Review of the Global Literature." Unpublished.
- Ramey, Garey, and Valerie A. Ramey.** 2010. "The Rug Rat Race." *Brookings Papers on Economic Activity* 40 (1): 129–76.
- Rossi, Pauline, and Yun Xiao.** 2024. "Spillovers in Childbearing Decisions and Fertility Transitions: Evidence from China." *Journal of the European Economic Association* 22 (1): 161–99.
- Schoonbroodt, Alice, and Michele Tertilt.** 2014. "Property Rights and Efficiency in OLG Models with Endogenous Fertility." *Journal of Economic Theory* 150: 551–82.
- Seth, Michael J.** 2002. *Education Fever: Society, Politics, and the Pursuit of Schooling in South Korea*. Honolulu, HI: University of Hawaii Press.
- Sommer, Kamila.** 2016. "Fertility Choice in a Life Cycle Model with Idiosyncratic Uninsurable Earnings Risk." *Journal of Monetary Economics* 83: 27–38.
- Sorensen, Clark W.** 1994. "Success and Education in South Korea." *Comparative Education Review* 38 (1): 10–35.
- Spence, Michael.** 1973. "Job Market Signaling." *Quarterly Journal of Economics* 87 (3): 355–74.
- Statistics Korea.** 2009. "MicroData Integrated Service (MDIS): Time Use Survey." Statistics Korea. <https://mdis.kostat.go.kr/index.do> (accessed October 31, 2022).
- Statistics Korea.** 2013. "MicroData Integrated Service (MDIS): Private Education Expenditures Survey, 2009-2013." Statistics Korea. <https://mdis.kostat.go.kr/index.do> (accessed November 18, 2022).
- Statistics Korea.** 2018. "MicroData Integrated Service (MDIS): Private Education Expenditures Survey, 2014-2018." Statistics Korea. <https://mdis.kostat.go.kr/index.do> (accessed January 26, 2023).
- Statistics Korea.** 2019. "Private Education Participation Rate by School Level." Statistics Korea. https://kosis.kr/statHtml/statHtml.do?orgId=101&tblId=DT_1PE301&conn_path=I2&language=en (accessed April 30, 2021).
- Statistics Korea.** 2022. "Social Survey." Statistics Korea. <https://kostat.go.kr/menu.es?mid=a20205030000> (accessed February 27, 2023).

- Tauchen, George.** 1986. "Finite State Markov-Chain Approximations to Univariate and Vector Autoregressions." *Economics Letters* 20 (2): 177–81.
- Veblen, Thorstein.** 1899. *The Theory of the Leisure Class*. Hamilton, ON, Canada: McMaster University Archive for the History of Economic Thought.
- Wong, Nancy Y., and Aaron C. Ahuvia.** 1998. "Personal Taste and Family Face: Luxury Consumption in Confucian and Western Societies." *Psychology & Marketing* 15 (5): 423–41.
- Yum, Minchul.** 2023. "Parental Time Investment and Intergenerational Mobility." *International Economic Review* 64 (1): 187–223.